```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                     2 ************************************************************************
                                                     3 *
                                                     4 *        EXPERIMENTAL pending further PoP definition
                                                     5 *
                                                     6 *        Zvector E6 instruction tests for VRR-c encoded:
                                                     7 *
                                                     8 *        E675  VCRNF   - VECTOR FP CONVERT AND ROUND TO NNP
                                                     9 *
                                                    10 *        and partial testing of
                                                    11 *
                                                    12 *        E656 VCLFNH   - VECTOR FP CONVERT AND LENGTHEN FROM NNP HIGH
                                                    13 *        E65E VCLFNL   - VECTOR FP CONVERT AND LENGTHEN FROM NNP LOW
                                                    14 *
                                                    15 *        during cross check tests for VCRNF
                                                    16 *
                                                    17 *        James Wekel August 2024
                                                    18 ************************************************************************


                                                    20 ************************************************************************
                                                    21 *
                                                    22 *        basic instruction tests
                                                    23 *
                                                    24 ************************************************************************
                                                    25 *  This program tests EXPERIMENTAL functioning of the z/arch E6 VRR-c
                                                    26 *  Neural-network-processing-assist facility vector instructions.
                                                    27 *  These test are EXPERIMENTAL pending further PoP definition of
                                                    28 *  NNP-data-type-1.
                                                    29 *
                                                    30 *  If requested and if VXC == 0 after test instruction execution,
                                                    31 *  a cross check test is performed. A cross check uses the result
                                                    32 *  of the instruction test to recreate the test source.
                                                    33 *
                                                    34 *  Exceptions (including trapable IEEE exceptions) are not tested.
                                                    35 *
                                                    36 *  PLEASE NOTE that the tests are very SIMPLE TESTS designed to catch
                                                    37 *  obvious coding errors.  None of the tests are thorough.  They are
                                                    38 *  NOT designed to test all aspects of any of the instructions.
                                                    39 *
                                                    40 ************************************************************************
                                                    41 *
                                                    42 *   *Testcase zvector-e6-21-VCRNF
                                                    43 *   *
                                                    44 *   *        EXPERIMENTAL pending further PoP definition
                                                    45 *   *
                                                    46 *   *        Zvector E6 instruction tests for VRR-c encoded:
                                                    47 *   *
                                                    48 *   *        E675 VCRNF   - VECTOR FP CONVERT AND ROUND TO NNP
                                                    49 *   *
                                                    50 *   *        # -------------------------------------------------------
                                                    51 *   *        #  This tests only the basic function of the instruction.
                                                    52 *   *        #  Exceptions are NOT tested.
                                                    53 *   *        # -------------------------------------------------------
                                                    54 *   *
                                                    55 *   mainsize    2
                                                    56 *   numcpu      1
```

```
 LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                 *   sysclear
                                                 *   archlvl     z/Arch
                                                 *
                                                 *   loadcore    "$(testpath)/zvector-e6-21-VCRNF.core" 0x0
                                                 *
                                                 *   diag8cmd    enable    # (needed for messages to Hercules console)
                                                 *   runtest 5
                                                 *   diag8cmd    disable   # (reset back to default)
                                                 *
                                                 *   *Done
                                                 *
                                                 ******************************************************************
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                  70 ***********************************************************************
                                                  71 *        FCHECK Macro - Is a Facility Bit set?
                                                  72 *
                                                  73 *        If the facility bit is NOT set, an message is issued and
                                                  74 *        the test is skipped.
                                                  75 *
                                                  76 *        Fcheck uses R0, R1 and R2
                                                  77 *
                                                  78 * eg.    FCHECK 134,'vector-packed-decimal'
                                                  79 ***********************************************************************
                                                  80          MACRO
                                                  81          FCHECK &BITNO,&NOTSETMSG
                                                  82 .*                        &BITNO : facility bit number to check
                                                  83 .*                        &NOTSETMSG : 'facility name'
                                                  84          LCLA  &FBBYTE           Facility bit in Byte
                                                  85          LCLA  &FBBIT            Facility bit within Byte
                                                  86
                                                  87          LCLA  &L(8)
                                                  88 &L(1)    SetA  128,64,32,16,8,4,2,1  bit positions within byte
                                                  89
                                                  90 &FBBYTE  SETA  &BITNO/8
                                                  91 &FBBIT   SETA  &L((&BITNO-(&FBBYTE*8))+1)
                                                  92 .*       MNOTE 0,'checking Bit=&BITNO: FBBYTE=&FBBYTE, FBBIT=&FBBIT'
                                                  93
                                                  94          B     X&SYSNDX
                                                  95 *                                      Fcheck data area
                                                  96 *                                      skip messgae
                                                  97 SKT&SYSNDX DC  C'          Skipping tests: '
                                                  98          DC    C&NOTSETMSG
                                                  99          DC    C' facility (bit &BITNO) is not installed.'
                                                 100 SKL&SYSNDX EQU *-SKT&SYSNDX
                                                 101 *                                      facility bits
                                                 102          DS    FD                      gap
                                                 103 FB&SYSNDX DS   4FD
                                                 104          DS    FD                      gap
                                                 105 *
                                                 106 X&SYSNDX EQU *
                                                 107          LA    R0,((X&SYSNDX-FB&SYSNDX)/8)-1
                                                 108          STFLE FB&SYSNDX               get facility bits
                                                 109
                                                 110          XGR   R0,R0
                                                 111          IC    R0,FB&SYSNDX+&FBBYTE    get fbit byte
                                                 112          N     R0,=F'&FBBIT'           is bit set?
                                                 113          BNZ   XC&SYSNDX
                                                 114 *
                                                 115 * facility bit not set, issue message and exit
                                                 116 *
                                                 117          LA    R0,SKL&SYSNDX           message length
                                                 118          LA    R1,SKT&SYSNDX           message address
                                                 119          BAL   R2,MSG
                                                 120
                                                 121          B     EOJ
                                                 122 XC&SYSNDX EQU *
                                                 123          MEND
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                  125 ***********************************************************************
                                                  126 *        Low core PSWs
                                                  127 ***********************************************************************


                             00000000  00002D83   129 ZVE6TST  START 0
00000000                     00000000              130          USING ZVE6TST,R0            Low core addressability
                                                   131
                             00000140  00000000   132 SVOLDPSW EQU   ZVE6TST+X'140'        z/Arch Supervisor call old PSW


00000000                     00000000  000001A0   134          ORG   ZVE6TST+X'1A0'        z/Architecture RESTART PSW
000001A0  00000001 80000000                       135          DC    X'0000000180000000'
000001A8  00000000 00000200                       136          DC    AD(BEGIN)


000001B0                     000001B0  000001D0   138          ORG   ZVE6TST+X'1D0'        z/Architecture PROGRAM CHECK PSW
000001D0  00020001 80000000                       139          DC    X'0002000180000000'
000001D8  00000000 0000DEAD                       140          DC    AD(X'DEAD')


000001E0                     000001E0  00000200   142          ORG   ZVE6TST+X'200'        Start of actual test program...
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                    144
                                                    145 ************************************************************************
                                                    146 *                The actual "ZVE6TST" program itself...
                                                    147 ************************************************************************
                                                    148 *
                                                    149 *  Architecture Mode: z/Arch
                                                    150 *  Register Usage:
                                                    151 *
                                                    152 *   R0       (work)
                                                    153 *   R1-4     (work)
                                                    154 *   R5       Testing control table - current test base
                                                    155 *   R6       (work)
                                                    156 *   R7       msg return
                                                    157 *   R8       First base register
                                                    158 *   R9       Second base register
                                                    159 *   R10      Third base register
                                                    160 *   R11      E6TEST call return
                                                    161 *   R12      E6TESTS register
                                                    162 *   R13      (work)
                                                    163 *   R14      Subroutine call
                                                    164 *   R15      Secondary Subroutine call or work
                                                    165 *
                                                    166 ************************************************************************

00000200                       00000200             168          USING  BEGIN,R8               FIRST Base Register
00000200                       00001200             169          USING  BEGIN+4096,R9          SECOND Base Register
00000200                       00002200             170          USING  BEGIN+8192,R10         THIRD Base Register
                                                    171
00000200  0580                                      172 BEGIN    BALR  R8,0                    Initialize FIRST base register
00000202  0680                                      173          BCTR  R8,0                    Initialize FIRST base register
00000204  0680                                      174          BCTR  R8,0                    Initialize FIRST base register
                                                    175
00000206  4190 8800                     00000800    176          LA    R9,2048(,R8)            Initialize SECOND base register
0000020A  4190 9800                     00000800    177          LA    R9,2048(,R9)            Initialize SECOND base register
                                                    178
0000020E  41A0 9800                     00000800    179          LA    R10,2048(,R9)           Initialize THIRD base register
00000212  41A0 A800                     00000800    180          LA    R10,2048(,R10)          Initialize THIRD base register
                                                    181
00000216  B600 8524                     00000724    182          STCTL R0,R0,CTLR0             Store CR0 to enable AFP
0000021A  9604 8525                     00000725    183          OI    CTLR0+1,X'04'           Turn on AFP bit
0000021E  9602 8525                     00000725    184          OI    CTLR0+1,X'02'           Turn on Vector bit
00000222  B700 8524                     00000724    185          LCTL  R0,R0,CTLR0             Reload updated CR0
                                                    186
                                                    187 ************************************************************************
                                                    188 * Is Neural-network-processing-assist facility 2 installed  (bit 165)
                                                    189 ************************************************************************
                                                    190
                                                    191          FCHECK 165,'Neural-network-processing-assist'
00000226  47F0 80C0                     000002C0    192+         B     X0001
                                                    193+*                                       Fcheck data area
                                                    194+*                                       skip messgae
0000022A  40404040 40404040                         195+SKT0001  DC    C'          Skipping tests: '
00000244  D585A499 81936095                         196+         DC    C'Neural-network-processing-assist'
00000264  40868183 899389A3                         197+         DC    C' facility (bit 165) is not installed.'
                               0000005F  00000001   198+SKL0001  EQU   *-SKT0001
                                                    199+*                                       facility bits
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00000290  00000000 00000000                        200+         DS    FD                      gap
00000298  00000000 00000000                        201+FB0001   DS    4FD
000002B8  00000000 00000000                        202+         DS    FD                      gap
                                                   203+*
                             000002C0  00000001    204+X0001    EQU   *
000002C0  4100 0004                    00000004    205+         LA    R0,((X0001-FB0001)/8)-1
000002C4  B2B0 8098                    00000298    206+         STFLE FB0001                  get facility bits
000002C8  B982 0000                                207+         XGR   R0,R0
000002CC  4300 80AC                    000002AC    208+         IC    R0,FB0001+20            get fbit byte
000002D0  5400 8560                    00000760    209+         N     R0,=F'4'                is bit set?
000002D4  4770 80E8                    000002E8    210+         BNZ   XC0001
                                                   211+*
                                                   212+* facility bit not set, issue message and exit
                                                   213+*
000002D8  4100 005F                    0000005F    214+         LA    R0,SKL0001              message length
000002DC  4110 802A                    0000022A    215+         LA    R1,SKT0001              message address
000002E0  4520 8438                    00000638    216+         BAL   R2,MSG
000002E4  47F0 8508                    00000708    217+         B     EOJ
                             000002E8  00000001    218+XC0001   EQU   *
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                  ***********************************************************************
                                                  *              Do tests in the E6TESTS table
                                                  ***********************************************************************

000002E8  58C0 8564                    00000764          L     R12,=A(E6TESTS)        get table of test addresses

                             000002EC  00000001  NEXTE6   EQU   *
000002EC  5850 C000                    00000000          L     R5,0(0,R12)            get test address
000002F0  1255                                           LTR   R5,R5                     have a test?
000002F2  4780 83F4                    000005F4          BZ    ENDTEST                      done?

000002F6                     00000000                    USING E6TEST,R5

000002F6  4800 5004                    00000004          LH    R0,TNUM                save current test number
000002FA  5000 8E04                    00001004          ST    R0,TESTING             for easy reference

000002FE  58B0 5000                    00000000          L     R11,TSUB               get address of test routine
00000302  05BB                                           BALR  R11,R11                do test

00000304  45F0 814E                    0000034E          BAL   R15,XCHECK
                                                  *
                                                  * validate FPC first
                                                  *
00000308  D500 500A 5041               00000041          CLC   FLG(1),FPC_R+1          expected FPC flags?
0000030E  4780 8116                    00000316          BE    CHECKVXC
00000312  4570 8290                    00000490          BAL   R7,FPCMSG               no, issue failed message

                             00000316  00000001  CHECKVXC EQU   *
00000316  D500 500B 5042               00000042          CLC   VXC(1),FPC_R+2          expected VXC?
0000031C  4780 8124                    00000324          BE    CHECKRES
00000320  4570 8302                    00000502          BAL   R7,VXCMSG               no, issue failed message

                             00000324  00000001  CHECKRES EQU   *

                                                  * then validate results, if not inexact

00000324  B982 0011                                      XGR   R1,R1
00000328  4310 5041                    00000041          IC    R1,FPC_R+1             FPC flags
0000032C  5410 8568                    00000768          N     R1,=XL4'00000008'      check inexact flag
00000330  1211                                           LTR   R1,R1
00000332  4770 8146                    00000346          BNZ   DONEXT

00000336  E310 501C 0014               0000001C          LGF   R1,READDR              expected result address
0000033C  D50F 5028 1000     00000028  00000000          CLC   V1OUTPUT,0(R1)
00000342  4770 8374                    00000574          BNE   FAILMSG                no, issue failed message

                             00000346  00000001  DONEXT   EQU   *
00000346  41C0 C004                    00000004          LA    R12,4(0,R12)           next test address
0000034A  47F0 80EC                    000002EC          B     NEXTE6
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   270 *-----------------------------------------------------------------------
                                                   271 * cross check that the result can be converted back to the source
                                                   272 * where there were NO exceptions (DXC/VXC = 0)
                                                   273 *-----------------------------------------------------------------------
                                0000034E  00000001 274 XCHECK   EQU   *
0000034E  D207 5050 8548        00000050  00000748 275          MVC   SKIPXC,=CL8'SKIP XC '
00000354  D500 5007 8574        00000007  00000774 276          CLC   XCSKIP,=CL1'S'         skip xcheck requested
0000035A  478F 0000                       00000000 277          BE    0(R15)                   skip = S, so exit
0000035E  9500 5042                       00000042 278          CLI   FPC_R+2,0              Only Xcheck no VXC error
00000362  477F 0000                       00000000 279          BNZ   0(R15)                   some VXC error , so exit
                                                   280 *
                                                   281 *        cross check depends on the instruction
                                                   282 *
00000366  D207 5050 8550        00000050  00000750 283          MVC   SKIPXC,=CL8'        '
0000036C  D507 5010 8558        00000010  00000758 284          CLC   OPNAME,=CL8'VCRNF'
00000372  4780 8180                       00000380 285          BE    XCVCRNF
                                                   286
                                                   287
00000376  D207 5050 8548        00000050  00000748 288          MVC   SKIPXC,=CL8'SKIP XC '
0000037C  07FF                                     289          BR    R15                    return from xcheck
                                                   290
                                                   291 * cross check: VCRNF   - VECTOR FP CONVERT AND ROUND TO NNP
                                                   292 *
00000380                                           293 XCVCRNF  DS    0F
00000380  B29D 852C                       0000072C 294          LFPC  FPCMOFF                initialize FPC
                                                   295
00000384  E700 5028 0806                  00000028 296          VL    V16,V1OUTPUT
0000038A  E6F0 0000 2456                           297          VCLFNH V15,V16,2,0
00000390  B29C 5088                       00000088 298          STFPC FPC_XC1                save 1st FPC
00000394  E7F0 5060 000E                  00000060 299          VST   V15,XCOUTPUT
                                                   300
0000039A  B29D 852C                       0000072C 301          LFPC  FPCMOFF                initialize FPC
                                                   302
0000039E  E700 5028 0806                  00000028 303          VL    V16,V1OUTPUT
000003A4  E6F0 0000 245E                           304          VCLFNL V15,V16,2,0
000003AA  B29C 508C                       0000008C 305          STFPC FPC_XC2                save 2nd FPC
000003AE  E7F0 5070 000E                  00000070 306          VST   V15,XCOUTPUT+16
                                                   307
000003B4  9500 508A                       0000008A 308          CLI   FPC_XC1+2,0            1st FPC
000003B8  477F 0000                       00000000 309          BNZ   0(R15)                 some VXC error , so exit
                                                   310
000003BC  9500 508E                       0000008E 311          CLI   FPC_XC2+2,0            2nd FPC
000003C0  477F 0000                       00000000 312          BNZ   0(R15)                 some VXC error , so exit
                                                   313
000003C4  E310 500C 0014                  0000000C 314          LGF   R1,V2ADDR              expected source address
000003CA  D51F 5060 1000                  00000000 315          CLC   XCOUTPUT(32),0(R1)
000003D0  4770 81D6                       000003D6 316          BNE   XCFAILMSG
000003D4  07FF                                     317          BR    R15                    return from xcheck
                                                   318
                                                   319
                                                   320 * xcheck failed message
000003D6                                           321 XCFAILMSG DS   0H
000003D6  5800 856C                       0000076C 322          L     R0,=F'1'               set failed test indicator
000003DA  5000 8E00                       00001000 323          ST    R0,FAILED
                                                   324
000003DE  4820 5004                       00000004 325          LH    R2,TNUM                get test number and convert
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

000003E2  4E20 8F90                    00001190   326          CVD   R2,DECNUM
000003E6  D211 8F7A 8F64      0000117A 00001164   327          MVC   PRT3,EDIT
000003EC  DE11 8F7A 8F90      0000117A 00001190   328          ED    PRT3,DECNUM
000003F2  D202 8F23 8F87      00001123 00001187   329          MVC   XCPTNUM(3),PRT3+13     fill in message with test #
                                                  330
000003F8  D207 8F45 5010      00001145 00000010   331          MVC   XCPNAME,OPNAME         fill in message with instruction
                                                  332
000003FE  B982 0022                               333          XGR   R2,R2
00000402  4320 5008                    00000008   334          IC    R2,M4                  get m3 and convert
00000406  4E20 8F90                    00001190   335          CVD   R2,DECNUM
0000040A  D211 8F7A 8F64      0000117A 00001164   336          MVC   PRT3,EDIT
00000410  DE11 8F7A 8F90      0000117A 00001190   337          ED    PRT3,DECNUM
00000416  D201 8F56 8F88      00001156 00001188   338          MVC   XCPM4(2),PRT3+14       fill in message with m3 field
                                                  339 *
0000041C  B982 0022                               340          XGR   R2,R2
00000420  4320 5009                    00000009   341          IC    R2,M5                  get m4 and convert
00000424  4E20 8F90                    00001190   342          CVD   R2,DECNUM
00000428  D211 8F7A 8F64      0000117A 00001164   343          MVC   PRT3,EDIT
0000042E  DE11 8F7A 8F90      0000117A 00001190   344          ED    PRT3,DECNUM
00000434  D201 8F61 8F88      00001161 00001188   345          MVC   XCPM5(2),PRT3+14       fill in message with m4 field
                                                  346
0000043A  50F0 8288                    00000488   347          ST    R15,XCR15              save r15
0000043E  4100 004E                    0000004E   348          LA    R0,XCPLNG              message length
00000442  4110 8F16                    00001116   349          LA    R1,XCPLINE             message address
00000446  45F0 8402                    00000602   350          BAL   R15,RPTERROR
0000044A  58F0 8288                    00000488   351          L     R15,XCR15
                                                  352
0000044E  5800 856C                    0000076C   353          L     R0,=F'1'               set failed test indicator
00000452  5000 8E00                    00001000   354          ST    R0,FAILED
00000456  07FF                                    355          BR    R15                    return from xcheck
                                                  356
00000458                                          357          DS    0FD
00000458  00000000 00000000                       358 XCRESULT DS    XL16
00000468  00000000 00000000                       359 XCV1     DS    XL16
00000478  00000000 00000000                       360 XCV2     DS    XL16
00000488  00000000 00000000                       361 XCR15    DS    FD
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                  363 ***********************************************************************
                                                  364 * FPC not as expected:
                                                  365 *        issue message with test number, instruction under test,
                                                  366 *              expected FPC and received FPC
                                                  367 ***********************************************************************
                              00000490  00000001  368 FPCMSG   EQU   *
00000490  5800 856C                     0000076C  369          L     R0,=F'1'                set failed test indicator
00000494  5000 8E00                     00001000  370          ST    R0,FAILED
                                                  371
00000498  4820 5004                     00000004  372          LH    R2,TNUM                 get test number and convert
0000049C  4E20 8F90                     00001190  373          CVD   R2,DECNUM
000004A0  D211 8F7A 8F64      0000117A  00001164  374          MVC   PRT3,EDIT
000004A6  DE11 8F7A 8F90      0000117A  00001190  375          ED    PRT3,DECNUM
000004AC  D202 8E15 8F87      00001015  00001187  376          MVC   FPCPNUM(3),PRT3+13      fill in message with test #
                                                  377
000004B2  D207 8E39 5010      00001039  00000010  378          MVC   FPCPNAME,OPNAME         fill in message with instruction
                                                  379 *
000004B8  B982 0022                               380          XGR   R2,R2
000004BC  4320 500A                     0000000A  381          IC    R2,FLG                  get expected flags and convert
000004C0  4E20 8F90                     00001190  382          CVD   R2,DECNUM
000004C4  D211 8F7A 8F64      0000117A  00001164  383          MVC   PRT3,EDIT
000004CA  DE11 8F7A 8F90      0000117A  00001190  384          ED    PRT3,DECNUM
000004D0  D202 8E52 8F87      00001052  00001187  385          MVC   FPCPEXP(3),PRT3+13      fill in message with m3 field
                                                  386 *
000004D6  B982 0022                               387          XGR   R2,R2
000004DA  4320 5041                     00000041  388          IC    R2,FPC_R+1              get received flags and convert
000004DE  4E20 8F90                     00001190  389          CVD   R2,DECNUM
000004E2  D211 8F7A 8F64      0000117A  00001164  390          MVC   PRT3,EDIT
000004E8  DE11 8F7A 8F90      0000117A  00001190  391          ED    PRT3,DECNUM
000004EE  D202 8E67 8F87      00001067  00001187  392          MVC   FPCPGOT(3),PRT3+13      fill in message with FPC got
                                                  393 *
000004F4  4100 0063                     00000063  394          LA    R0,FPCPLNG              message length
000004F8  4110 8E08                     00001008  395          LA    R1,FPCPLINE             message address
000004FC  45F0 8402                     00000602  396          BAL   R15,RPTERROR
00000500  07F7                                    397          BR    R7
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                  399 ***********************************************************************
                                                  400 * VXC not as expected:
                                                  401 *        issue message with test number, instruction under test,
                                                  402 *              expected FPC and received FPC
                                                  403 ***********************************************************************
                              00000502  00000001  404 VXCMSG   EQU   *
00000502  5800 856C                     0000076C  405          L     R0,=F'1'                set failed test indicator
00000506  5000 8E00                     00001000  406          ST    R0,FAILED
                                                  407
0000050A  4820 5004                     00000004  408          LH    R2,TNUM                 get test number and convert
0000050E  4E20 8F90                     00001190  409          CVD   R2,DECNUM
00000512  D211 8F7A 8F64      0000117A  00001164  410          MVC   PRT3,EDIT
00000518  DE11 8F7A 8F90      0000117A  00001190  411          ED    PRT3,DECNUM
0000051E  D202 8E78 8F87      00001078  00001187  412          MVC   VXCPNUM(3),PRT3+13      fill in message with test #
                                                  413
00000524  D207 8E9C 5010      0000109C  00000010  414          MVC   VXCPNAME,OPNAME         fill in message with instruction
                                                  415 *
0000052A  B982 0022                               416          XGR   R2,R2
0000052E  4320 500B                     0000000B  417          IC    R2,VXC                  get expected flags and convert
00000532  4E20 8F90                     00001190  418          CVD   R2,DECNUM
00000536  D211 8F7A 8F64      0000117A  00001164  419          MVC   PRT3,EDIT
0000053C  DE11 8F7A 8F90      0000117A  00001190  420          ED    PRT3,DECNUM
00000542  D202 8EB5 8F87      000010B5  00001187  421          MVC   VXCPEXP(3),PRT3+13      fill in message with m3 field
                                                  422 *
00000548  B982 0022                               423          XGR   R2,R2
0000054C  4320 5042                     00000042  424          IC    R2,FPC_R+2              get received flags and convert
00000550  4E20 8F90                     00001190  425          CVD   R2,DECNUM
00000554  D211 8F7A 8F64      0000117A  00001164  426          MVC   PRT3,EDIT
0000055A  DE11 8F7A 8F90      0000117A  00001190  427          ED    PRT3,DECNUM
00000560  D202 8ECA 8F87      000010CA  00001187  428          MVC   VXCPGOT(3),PRT3+13      fill in message with FPC got
                                                  429 *
00000566  4100 0063                     00000063  430          LA    R0,VXCPLNG              message length
0000056A  4110 8E6B                     0000106B  431          LA    R1,VXCPLINE             message address
0000056E  45F0 8402                     00000602  432          BAL   R15,RPTERROR
00000572  07F7                                    433          BR    R7
                                                  434
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                  436 ***********************************************************************
                                                  437 * result not as expected:
                                                  438 *        issue message with test number, instruction under test
                                                  439 *              and instruction m4
                                                  440 ***********************************************************************
                              00000574  00000001  441 FAILMSG  EQU   *
00000574  5800 856C                     0000076C  442          L     R0,=F'1'                set failed test indicator
00000578  5000 8E00                     00001000  443          ST    R0,FAILED
                                                  444
0000057C  4820 5004                     00000004  445          LH    R2,TNUM                 get test number and convert
00000580  4E20 8F90                     00001190  446          CVD   R2,DECNUM
00000584  D211 8F7A 8F64      0000117A  00001164  447          MVC   PRT3,EDIT
0000058A  DE11 8F7A 8F90      0000117A  00001190  448          ED    PRT3,DECNUM
00000590  D202 8EDB 8F87      000010DB  00001187  449          MVC   PRTNUM(3),PRT3+13       fill in message with test #
                                                  450
00000596  D207 8EF6 5010      000010F6  00000010  451          MVC   PRTNAME,OPNAME          fill in message with instruction
                                                  452 *
0000059C  B982 0022                               453          XGR   R2,R2
000005A0  4320 5008                     00000008  454          IC    R2,M4                   get m3 and convert
000005A4  4E20 8F90                     00001190  455          CVD   R2,DECNUM
000005A8  D211 8F7A 8F64      0000117A  00001164  456          MVC   PRT3,EDIT
000005AE  DE11 8F7A 8F90      0000117A  00001190  457          ED    PRT3,DECNUM
000005B4  D201 8F07 8F88      00001107  00001188  458          MVC   PRTM4(2),PRT3+14        fill in message with m3 field
                                                  459 *
000005BA  B982 0022                               460          XGR   R2,R2
000005BE  4320 5009                     00000009  461          IC    R2,M5                   get m4 and convert
000005C2  4E20 8F90                     00001190  462          CVD   R2,DECNUM
000005C6  D211 8F7A 8F64      0000117A  00001164  463          MVC   PRT3,EDIT
000005CC  DE11 8F7A 8F90      0000117A  00001190  464          ED    PRT3,DECNUM
000005D2  D201 8F13 8F88      00001113  00001188  465          MVC   PRTM5(2),PRT3+14        fill in message with m4 field
                                                  466 *
000005D8  4100 0048                     00000048  467          LA    R0,PRTLNG               message length
000005DC  4110 8ECE                     000010CE  468          LA    R1,PRTLINE              message address
000005E0  45F0 8402                     00000602  469          BAL   R15,RPTERROR


                                                  471 ***********************************************************************
                                                  472 * continue after a failed test
                                                  473 ***********************************************************************
                              000005E4  00000001  474 FAILCONT EQU   *
000005E4  5800 856C                     0000076C  475          L     R0,=F'1'                set failed test indicator
000005E8  5000 8E00                     00001000  476          ST    R0,FAILED
                                                  477
000005EC  41C0 C004                     00000004  478          LA    R12,4(0,R12)            next test address
000005F0  47F0 80EC                     000002EC  479          B     NEXTE6


                                                  481 ***********************************************************************
                                                  482 * end of testing; set ending psw
                                                  483 ***********************************************************************
                              000005F4  00000001  484 ENDTEST  EQU   *
000005F4  5810 8E00                     00001000  485          L     R1,FAILED               did a test fail?
000005F8  1211                                    486          LTR   R1,R1
000005FA  4780 8508                     00000708  487          BZ    EOJ                      No, exit
000005FE  47F0 8520                     00000720  488          B     FAILTEST                 Yes, exit with BAD PSW
                                                  489
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   491 ***********************************************************************
                                                   492 *        RPTERROR          Report instruction test in error
                                                   493 *                              R0 = MESSGAE LENGTH
                                                   494 *                              R1 = ADDRESS OF MESSAGE
                                                   495 ***********************************************************************

00000602  50F0 8420                     00000620   497 RPTERROR ST    R15,RPTSAVE            Save return address
00000606  5050 8424                     00000624   498          ST    R5,RPTSVR5             Save R5
                                                   499 *
                                                   500 *        Use Hercules Diagnose for Message to console
                                                   501 *
0000060A  9002 8428                     00000628   502          STM   R0,R2,RPTDWSAV         save regs used by MSG
0000060E  4520 8438                     00000638   503          BAL   R2,MSG                 call Hercules console MSG display
00000612  9802 8428                     00000628   504          LM    R0,R2,RPTDWSAV         restore regs


00000616  5850 8424                     00000624   506          L     R5,RPTSVR5             Restore R5
0000061A  58F0 8420                     00000620   507          L     R15,RPTSAVE            Restore return address
0000061E  07FF                                     508          BR    R15                    Return to caller

00000620  00000000                                 510 RPTSAVE  DC    F'0'                   R15 save area
00000624  00000000                                 511 RPTSVR5  DC    F'0'                   R5 save area

00000628  00000000 00000000                        513 RPTDWSAV DC    2D'0'                  R0-R2 save area for MSG call
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                 515 ***********************************************************************
                                                 516 *        Issue HERCULES MESSAGE pointed to by R1, length in R0
                                                 517 *              R2 = return address
                                                 518 ***********************************************************************

00000638  4900 8570                    00000770  520 MSG      CH    R0,=H'0'                Do we even HAVE a message?
0000063C  07D2                                   521          BNHR  R2                      No, ignore

0000063E  9002 8474                    00000674  523          STM   R0,R2,MSGSAVE           Save registers

00000642  4900 8572                    00000772  525          CH    R0,=AL2(L'MSGMSG)       Message length within limits?
00000646  47D0 844E                    0000064E  526          BNH   MSGOK                   Yes, continue
0000064A  4100 0064                    00000064  527          LA    R0,L'MSGMSG             No, set to maximum

0000064E  1820                                   529 MSGOK    LR    R2,R0                   Copy length to work register
00000650  0620                                   530          BCTR  R2,0                    Minus-1 for execute
00000652  4420 8480                    00000680  531          EX    R2,MSGMVC               Copy message to O/P buffer

00000656  4120 200A                    0000000A  533          LA    R2,1+L'MSGCMD(,R2)      Calculate true command length
0000065A  4110 8486                    00000686  534          LA    R1,MSGCMD               Point to true command

0000065E  83120008                               536          DC    X'83',X'12',X'0008'     Issue Hercules Diagnose X'008'
00000662  4780 846E                    0000066E  537          BZ    MSGRET                  Return if successful
                                                 538
00000666  1222                                   539          LTR   R2,R2                   Is Diag8 Ry (R2) 0?
00000668  4780 846E                    0000066E  540          BZ    MSGRET                    an error occurred but coninue
                                                 541
0000066C  0000                                   542          DC    H'0'                    CRASH for debugging purposes

0000066E  9802 8474                    00000674  544 MSGRET   LM    R0,R2,MSGSAVE           Restore registers
00000672  07F2                                   545          BR    R2                      Return to caller



00000674  00000000 00000000                      547 MSGSAVE  DC    3F'0'                   Registers save area
00000680  D200 848F 1000      0000068F  00000000  548 MSGMVC   MVC   MSGMSG(0),0(R1)         Executed instruction

00000686  D4E2C7D5 D6C8405C                      550 MSGCMD   DC    C'MSGNOH * '            *** HERCULES MESSAGE COMMAND ***
0000068F  40404040 40404040                      551 MSGMSG   DC    CL100' '                 The message text to be displayed
                                                 552
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                   554 ************************************************************************
                                                   555 *        Normal completion or Abnormal termination PSWs
                                                   556 ************************************************************************


000006F8  00020001 80000000                        558 EOJPSW   DC    0D'0',X'0002000180000000',AD(0)

00000708  B2B2 84F8                     000006F8   560 EOJ      LPSWE EOJPSW                 Normal completion


00000710  00020001 80000000                        562 FAILPSW  DC    0D'0',X'0002000180000000',AD(X'BAD')

00000720  B2B2 8510                     00000710   564 FAILTEST LPSWE FAILPSW                Abnormal termination


                                                   566 ************************************************************************
                                                   567 *        Working Storage
                                                   568 ************************************************************************


00000724  00000000                                 570 CTLR0    DS    F                      CR0
00000728  00000000                                 571          DS    F
0000072C  00000000                                 572 FPCMOFF  DC    XL4'00000000'          FPC masks off
00000730  F8000000                                 573 FPCMON   DC    XL4'F8000000'          FPC masks on
00000734  80000000                                 574 FPCMIO   DC    XL4'80000000'          FPC mask invalid operation
00000738  20000000                                 575 FPCMOV   DC    XL4'20000000'          FPC mask overflow
0000073C  10000000                                 576 FPCMUN   DC    XL4'10000000'          FPC mask underflow
00000740  08000000                                 577 FPCMIE   DC    XL4'08000000'          FPC mask inexact


                                                   579
00000748                                           580          LTORG ,                Literals pool
00000748  E2D2C9D7 40E7C340                        581                =CL8'SKIP XC '
00000750  40404040 40404040                        582                =CL8'        '
00000758  E5C3D9D5 C6404040                        583                =CL8'VCRNF'
00000760  00000004                                 584                =F'4'
00000764  00002D04                                 585                =A(E6TESTS)
00000768  00000008                                 586                =XL4'00000008'
0000076C  00000001                                 587                =F'1'
00000770  0000                                     588                =H'0'
00000772  0064                                     589                =AL2(L'MSGMSG)
00000774  E2                                       590                =CL1'S'
                                                   591
                                                   592 *        some constants
                                                   593
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                              00000400  00000001   594 K        EQU   1024             One KB
                              00001000  00000001   595 PAGE     EQU   (4*K)            Size of one page
                              00010000  00000001   596 K64      EQU   (64*K)           64 KB
                              00100000  00000001   597 MB       EQU   (K*K)             1 MB
                                                   598
                                                   599
                              AABBCCDD  00000001   600 REG2PATT EQU   X'AABBCCDD'    Polluted Register pattern
                              000000DD  00000001   601 REG2LOW  EQU         X'DD'    (last byte above)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                    603 *=====================================================================
                                                    604 *
                                                    605 *  NOTE: start data on an address that is easy to display
                                                    606 *        within Hercules
                                                    607 *
                                                    608 *=====================================================================
                                                    609
00000775                      00000775  00001000    610          ORG   ZVE6TST+X'1000'
00001000  00000000                                  611 FAILED   DC    F'0'                    some test failed?
00001004  00000000                                  612 TESTING  DC    F'0'                    current test #


                                                    614
                                                    615 ************************************************************************
                                                    616 *        TEST failed : FPC message
                                                    617 ************************************************************************
                                                    618 *
                                                    619 *        failed message and associated editting
                                                    620 *
00001008  40404040 4040E385                         621 FPCPLINE DC    C'      Test # '
00001015  A7A7A7                                    622 FPCPNUM  DC    C'xxx'
00001018  40A69996 958740C6                         623          DC    c' wrong FPC flags for instruction '
00001039  A7A7A7A7 A7A7A7A7                         624 FPCPNAME DC    CL8'xxxxxxxx'
00001041  4085A797 8583A385                         625          DC    C' expected: flags='
00001052  A7A7A7                                    626 FPCPEXP  DC    C'xxx'
00001055  6B                                        627          DC    C','
00001056  40998583 8589A585                         628          DC    C' received: flags='
00001067  A7A7A7                                    629 FPCPGOT  DC    C'xxx'
0000106A  4B                                        630          DC    C'.'
                              00000063  00000001    631 FPCPLNG  EQU   *-FPCPLINE


                                                    633
                                                    634 ************************************************************************
                                                    635 *        TEST failed : VXC message
                                                    636 ************************************************************************
                                                    637 *
                                                    638 *        failed message and associated editting
                                                    639 *
0000106B  40404040 4040E385                         640 VXCPLINE DC    C'      Test # '
00001078  A7A7A7                                    641 VXCPNUM  DC    C'xxx'
0000107B  40A69996 958740C6                         642          DC    c' wrong FPC VXC   for instruction '
0000109C  A7A7A7A7 A7A7A7A7                         643 VXCPNAME DC    CL8'xxxxxxxx'
000010A4  4085A797 8583A385                         644          DC    C' expected:   VXC='
000010B5  A7A7A7                                    645 VXCPEXP  DC    C'xxx'
000010B8  6B                                        646          DC    C','
000010B9  40998583 8589A585                         647          DC    C' received:   VXC='
000010CA  A7A7A7                                    648 VXCPGOT  DC    C'xxx'
000010CD  4B                                        649          DC    C'.'
                              00000063  00000001    650 VXCPLNG  EQU   *-VXCPLINE
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                    652
                                                    653 ************************************************************************
                                                    654 *        TEST failed : result messgae
                                                    655 ************************************************************************
                                                    656 *
                                                    657 *        failed message and associated editting
                                                    658 *
000010CE  40404040 4040E385                         659 PRTLINE  DC    C'      Test # '
000010DB  A7A7A7                                    660 PRTNUM   DC    C'xxx'
000010DE  40868189 93858440                         661          DC    c' failed for instruction '
000010F6  A7A7A7A7 A7A7A7A7                         662 PRTNAME  DC    CL8'xxxxxxxx'
000010FE  40A689A3 884094F4                         663          DC    C' with m4='
00001107  A7A7                                      664 PRTM4    DC    C'xx'
00001109  6B                                        665          DC    C','
0000110A  40A689A3 884094F5                         666          DC    C' with m5='
00001113  A7A7                                      667 PRTM5    DC    C'xx'
00001115  4B                                        668          DC    C'.'
                            00000048  00000001      669 PRTLNG   EQU   *-PRTLINE


                                                    671 ************************************************************************
                                                    672 *        TEST failed : XCHECK
                                                    673 ************************************************************************
                                                    674 *
                                                    675 *        XCHECK failed message
                                                    676 *
00001116  40404040 4040E385                         677 XCPLINE  DC    C'      Test # '
00001123  A7A7A7                                    678 XCPTNUM  DC    C'xxx'
00001126  40E7C3C8 C5C3D240                         679          DC    c' XCHECK failed for instruction '
00001145  A7A7A7A7 A7A7A7A7                         680 XCPNAME  DC    CL8'xxxxxxxx'
0000114D  40A689A3 884094F4                         681          DC    C' with m4='
00001156  A7A7                                      682 XCPM4    DC    C'xx'
00001158  40A689A3 884094F5                         683          DC    C' with m5='
00001161  A7A7                                      684 XCPM5    DC    C'xx'
00001163  4B                                        685          DC    C'.'
                            0000004E  00000001      686 XCPLNG   EQU   *-XCPLINE
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                    688 ***********************************************************************
                                                    689 *        TEST failed : message working storge
                                                    690 ***********************************************************************
00001164  40212020 20202020                         691 EDIT     DC    XL18'402120202020202020202020202020202020'
                                                    692
00001176  7E7E7E6E                                  693          DC    C'===>'
0000117A  40404040 40404040                         694 PRT3     DC    CL18' '
0000118C  4C7E7E7E                                  695          DC    C'<===='
00001190  00000000 00000000                         696 DECNUM   DS    CL16


                                                    698 *
                                                    699 *        Vector instruction results, pollution and input
                                                    700 *
000011A0  00000000 00000000                         701          DS    XL16                                          gap
000011B0  FFFFFFFF FFFFFFFF                         702 V1FUDGE  DC    XL16'FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF'    V1 FUDGE
000011C0  12345678 90123456                         703 V1INPUT  DC    XL16'12345678901234567890123456789 01D'   V1 input
000011D0  00000000 00000000                         704          DS    XL16
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                    706 ************************************************************************
                                                    707 *        E6TEST DSECT
                                                    708 ************************************************************************


                                                    710 E6TEST   DSECT ,
00000000  00000000                                  711 TSUB     DC    A(0)              pointer  to test
00000004  0000                                      712 TNUM     DC    H'00'             Test Number
00000006  00                                        713          DC    X'00'
00000007  40                                        714 XCSKIP   DC    CL1' '            Y = skip cross check
00000008  00                                        715 M4       DC    HL1'00'           m4 used
00000009  00                                        716 M5       DC    HL1'00'           m5 used
0000000A  00                                        717 FLG      DC    X'00'             expected FPC flags
0000000B  00                                        718 VXC      DC    X'00'             VXC expected
0000000C  00000000                                  719 V2ADDR   DC    A(0)              address of v2: 16-byte packed decimal
00000010  40404040 40404040                         720 OPNAME   DC    CL8' '            E6 name
00000018  00000000                                  721 RELEN    DC    A(0)              result length
0000001C  00000000                                  722 READDR   DC    A(0)              expected result address
00000020  00000000 00000000                         723          DS    FD                   gap
00000028  00000000 00000000                         724 V1OUTPUT DS    XL16              V1 Output
00000038  00000000 00000000                         725          DS    FD                   gap
00000040  00000000                                  726 FPC_R    DS    F                 FPC after instruction
00000048  00000000 00000000                         727          DS    FD                   gap
00000050  40404040 40404040                         728 SKIPXC   DC    CL8' '            was cross check skipped?
00000058  00000000 00000000                         729          DS    FD                   gap
00000060  00000000 00000000                         730 XCOUTPUT DS    XL16              Cross check Output
00000070  00000000 00000000                         731          DS    XL16
00000080  00000000 00000000                         732          DS    FD                   gap
00000088  00000000                                  733 FPC_XC1  DS    F                 1st cross check FPC
0000008C  00000000                                  734 FPC_XC2  DS    F                 2nd cross check FPC
00000090  00000000 00000000                         735          DS    FD                   gap
00000098  00000000                                  736 WK1      DS    F                 debug area
0000009C  00000000                                  737 WK2      DS    F
000000A0                                            738          DS    0F
                                                    739 **
                                                    740 *        test routine will be here (from VRR-c macro)


                              00000000  00002D83    742 ZVE6TST  CSECT ,
000011E0                                            743          DS    0F


                                                    745 ************************************************************************
                                                    746 *     Macros to help build test tables
                                                    747 ************************************************************************


                                                    749 *
                                                    750 * macro to generate individual test
                                                    751 *
                                                    752          MACRO
                                                    753          VRR_C &INST,&M4,&M5,&FLAGS,&VXC,&SKIP
                                                    754 .*                             &INST  - VRR-c instruction under test
                                                    755 .*                             &m4    - m4 field
                                                    756 .*                             &m5    - m5 field
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                                  .*                                &flags - expected FPC flags (HEX)
                                                  .*                                &VXC   - expected VXC (HEX)
                                                  .*                                &SKIP  - S = skip cross check
                                                           GBLA  &TNUM
                                                  &TNUM    SETA  &TNUM+1

                                                           DS    0FD
                                                           USING *,R5               base for test data and test routine

                                                  T&TNUM   DC    A(X&TNUM)          address of test routine
                                                           DC    H'&TNUM'           test number
                                                           DC    X'00'
                                                           DC    CL1'&SKIP'         Y = skip cross check
                                                           DC    HL1'&M4'           m4
                                                           DC    HL1'&M5'           m5
                                                  FLG&TNUM DC    X'&FLAGS'          expected FPC flags
                                                  VXC&TNUM DC    X'&VXC'            expected VXC
                                                  V2_&TNUM DC    A(RE&TNUM+16)      address of v2: 16-byte packed decimal
                                                           DC    CL8'&INST'         instruction name
                                                           DC    A(16)              result length
                                                           DC    A(RE&TNUM)         address of expected resul
                                                           DS    FD                 gap
                                                  V1O&TNUM DS    XL16               V1 output
                                                           DS    FD                    gap
                                                  FPC_R_&TNUM DS F                  FPC after instruction
                                                           DS    FD                    gap
                                                           DC    CL8' '             was cross check skipped?
                                                           DS    FD                    gap
                                                  XCO&TNUM DS    XL16               Cross check Output
                                                           DS    XL16
                                                           DS    FD                    gap
                                                  FPC_XC_&TNUM DS F                1st cross check FPC
                                                           DS    F                 2nd cross check FPC
                                                           DS    FD                    gap
                                                           DS    F                  debug area
                                                           DS    F
                                                  .*
                                                  *
                                                  X&TNUM   DS    0F
                                                           LFPC  FPCMOFF            initialize FPC

                                                           LGF   R2,V2_&TNUM        get v2
                                                           VLM   V22,V23,0(R2)

                                                           &INST V22,V22,V23,&M4,&M5   test instruction (dest is source)

                                                           STFPC FPC_R_&TNUM        save FPC
                                                           VST   V22,V1O&TNUM       save instruction result

                                                           BR    R11                return

                                                  RE&TNUM  DS    0F                 expected 16 byte result
                                                           DROP  R5

                                                           MEND
```

```
 LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   813 *
                                                   814 * macro to generate table of pointers to individual tests
                                                   815 *
                                                   816          MACRO
                                                   817          PTTABLE
                                                   818          GBLA  &TNUM
                                                   819          LCLA  &CUR
                                                   820 &CUR     SETA  1
                                                   821 .*
                                                   822 TTABLE   DS    0F
                                                   823 .LOOP    ANOP
                                                   824 .*
                                                   825          DC    A(T&CUR)          TEST &CUR
                                                   826 .*
                                                   827 &CUR     SETA  &CUR+1
                                                   828          AIF   (&CUR LE &TNUM).LOOP
                                                   829 *
                                                   830          DC    A(0)              END OF TABLE
                                                   831          DC    A(0)
                                                   832 .*
                                                   833          MEND
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                    835 ************************************************************************
                                                    836 *        E6 VRR-c tests
                                                    837 ************************************************************************
                                                    838         PRINT DATA
                                                    839 *
                                                    840 *        E675  VCRNF   - VECTOR FP CONVERT AND ROUND TO NNP
                                                    841 *
                                                    842 *-----------------------------------------------------------------------
                                                    843 *   VRR-c   instruction, m4, m5, flags, VXC, SKIP
                                                    844 *           NOTE: flags and VXC are HEX values
                                                    845 *              and VXC should always by 00 as the FPC mask is zero
                                                    846 *   followed by
                                                    847 *              v1 - 16 byte expected result
                                                    848 *              v2 - 16 byte source
                                                    849 *              v3 - 16 byte source
                                                    850 *-----------------------------------------------------------------------
                                                    851 * VCRNF   - VECTOR FP CONVERT AND ROUND TO NNP
                                                    852 *-----------------------------------------------------------------------
                                                    853 * Short Float -> dlfloat  (with cross check dlfloat -> short float)
                                                    854 *
                                                    855 * some of these tests use numbers from PoP SA22-7832-13,
                                                    856 * Figure 9-2. Examples of Floating-Point Numbers ( page 9-6 )
                                                    857 *
                                                    858 * +0     simple instruction test and 'test' test
                                                    859         VRR_C VCRNF,0,2,00,00,N
000011E0                                            860+        DS    0FD
000011E0                       000011E0             861+        USING *,R5              base for test data and test routine
000011E0  00001280                                  862+T1      DC    A(X1)             address of test routine
000011E4  0001                                      863+        DC    H'1'              test number
000011E6  00                                        864+        DC    X'00'
000011E7  D5                                        865+        DC    CL1'N'            Y = skip cross check
000011E8  00                                        866+        DC    HL1'0'            m4
000011E9  02                                        867+        DC    HL1'2'            m5
000011EA  00                                        868+FLG1    DC    X'00'             expected FPC flags
000011EB  00                                        869+VXC1    DC    X'00'             expected VXC
000011EC  000012B4                                  870+V2_1    DC    A(RE1+16)         address of v2: 16-byte packed decimal
000011F0  E5C3D9D5 C6404040                         871+        DC    CL8'VCRNF'        instruction name
000011F8  00000010                                  872+        DC    A(16)             result length
000011FC  000012A4                                  873+        DC    A(RE1)            address of expected resul
00001200  00000000 00000000                         874+        DS    FD                gap
00001208  00000000 00000000                         875+V1O1    DS    XL16              V1 output
00001210  00000000 00000000
00001218  00000000 00000000                         876+        DS    FD                   gap
00001220  00000000                                  877+FPC_R_1 DS    F                 FPC after instruction
00001228  00000000 00000000                         878+        DS    FD                   gap
00001230  40404040 40404040                         879+        DC    CL8' '            was cross check skipped?
00001238  00000000 00000000                         880+        DS    FD                   gap
00001240  00000000 00000000                         881+XCO1    DS    XL16              Cross check Output
00001248  00000000 00000000
00001250  00000000 00000000                         882+        DS    XL16
00001258  00000000 00000000
00001260  00000000 00000000                         883+        DS    FD                   gap
00001268  00000000                                  884+FPC_XC_1 DS   F                1st cross check FPC
0000126C  00000000                                  885+        DS    F                2nd cross check FPC
00001270  00000000 00000000                         886+        DS    FD                   gap
00001278  00000000                                  887+        DS    F                 debug area
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

0000127C  00000000                                 888+         DS    F
                                                   889+*
00001280                                           890+X1       DS    0F
00001280  B29D 852C                     0000072C   891+         LFPC  FPCMOFF           initialize FPC
00001284  E320 500C 0014                000011EC   892+         LGF   R2,V2_1           get v2
0000128A  E767 2000 0C36                00000000   893+         VLM   V22,V23,0(R2)
00001290  E666 7002 0E75                           894+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
00001296  B29C 9020                     00001220   895+         STFPC FPC_R_1           save FPC
0000129A  E760 9008 080E                00001208   896+         VST   V22,V1O1          save instruction result
000012A0  07FB                                     897+         BR    R11               return
000012A4                                           898+RE1      DS    0F                expected 16 byte result
000012A4                                           899+         DROP  R5
000012A4  00000000 00000000                        900          DC    XL16'0000000000000000 0000000000000000'
000012AC  00000000 00000000
000012B4  00000000 00000000                        901          DC    XL16'0000000000000000 0000000000000000'
000012BC  00000000 00000000
000012C4  00000000 00000000                        902          DC    XL16'0000000000000000 0000000000000000'
000012CC  00000000 00000000
                                                   903
                                                   904 * +1, -1
                                                   905          VRR_C VCRNF,0,2,00,00,N
000012D8                                           906+         DS    0FD
000012D8                        000012D8           907+         USING *,R5              base for test data and test routine
000012D8  00001378                                 908+T2       DC    A(X2)             address of test routine
000012DC  0002                                     909+         DC    H'2'              test number
000012DE  00                                       910+         DC    X'00'
000012DF  D5                                       911+         DC    CL1'N'            Y = skip cross check
000012E0  00                                       912+         DC    HL1'0'            m4
000012E1  02                                       913+         DC    HL1'2'            m5
000012E2  00                                       914+FLG2     DC    X'00'             expected FPC flags
000012E3  00                                       915+VXC2     DC    X'00'             expected VXC
000012E4  000013AC                                 916+V2_2     DC    A(RE2+16)         address of v2: 16-byte packed decimal
000012E8  E5C3D9D5 C6404040                        917+         DC    CL8'VCRNF'        instruction name
000012F0  00000010                                 918+         DC    A(16)             result length
000012F4  0000139C                                 919+         DC    A(RE2)            address of expected resul
000012F8  00000000 00000000                        920+         DS    FD                gap
00001300  00000000 00000000                        921+V1O2     DS    XL16              V1 output
00001308  00000000 00000000
00001310  00000000 00000000                        922+         DS    FD                   gap
00001318  00000000                                 923+FPC_R_2  DS    F                 FPC after instruction
00001320  00000000 00000000                        924+         DS    FD                   gap
00001328  40404040 40404040                        925+         DC    CL8' '            was cross check skipped?
00001330  00000000 00000000                        926+         DS    FD                   gap
00001338  00000000 00000000                        927+XCO2     DS    XL16              Cross check Output
00001340  00000000 00000000
00001348  00000000 00000000                        928+         DS    XL16
00001350  00000000 00000000
00001358  00000000 00000000                        929+         DS    FD                   gap
00001360  00000000                                 930+FPC_XC_2 DS    F                1st cross check FPC
00001364  00000000                                 931+         DS    F                2nd cross check FPC
00001368  00000000 00000000                        932+         DS    FD                   gap
00001370  00000000                                 933+         DS    F                 debug area
00001374  00000000                                 934+         DS    F
                                                   935+*
00001378                                           936+X2       DS    0F
00001378  B29D 852C                     0000072C   937+         LFPC  FPCMOFF           initialize FPC
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

0000137C  E320 500C 0014                 000012E4   938+         LGF   R2,V2_2           get v2
00001382  E767 2000 0C36                 00000000   939+         VLM   V22,V23,0(R2)
00001388  E666 7002 0E75                            940+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
0000138E  B29C 5040                      00001318   941+         STFPC FPC_R_2           save FPC
00001392  E760 5028 080E                 00001300   942+         VST   V22,V1O2          save instruction result
00001398  07FB                                      943+         BR    R11               return
0000139C                                            944+RE2      DS    0F                expected 16 byte result
0000139C                                            945+         DROP  R5
0000139C  3E000000 00000000                         946          DC    XL16'3E00000000000000 BE00000000000000'
000013A4  BE000000 00000000
000013AC  3F800000 00000000                         947          DC    XL16'3F80000000000000 0000000000000000'
000013B4  00000000 00000000
000013BC  BF800000 00000000                         948          DC    XL16'BF80000000000000 0000000000000000'
000013C4  00000000 00000000
                                                    949
                                                    950 * +.5, -.5
                                                    951          VRR_C VCRNF,0,2,00,00,N
000013D0                                            952+         DS    0FD
000013D0                         000013D0           953+         USING *,R5              base for test data and test routine
000013D0  00001470                                  954+T3       DC    A(X3)             address of test routine
000013D4  0003                                      955+         DC    H'3'              test number
000013D6  00                                        956+         DC    X'00'
000013D7  D5                                        957+         DC    CL1'N'            Y = skip cross check
000013D8  00                                        958+         DC    HL1'0'            m4
000013D9  02                                        959+         DC    HL1'2'            m5
000013DA  00                                        960+FLG3     DC    X'00'             expected FPC flags
000013DB  00                                        961+VXC3     DC    X'00'             expected VXC
000013DC  000014A4                                  962+V2_3     DC    A(RE3+16)         address of v2: 16-byte packed decimal
000013E0  E5C3D9D5 C6404040                         963+         DC    CL8'VCRNF'        instruction name
000013E8  00000010                                  964+         DC    A(16)             result length
000013EC  00001494                                  965+         DC    A(RE3)            address of expected resul
000013F0  00000000 00000000                         966+         DS    FD                gap
000013F8  00000000 00000000                         967+V1O3     DS    XL16              V1 output
00001400  00000000 00000000
00001408  00000000 00000000                         968+         DS    FD                   gap
00001410  00000000                                  969+FPC_R_3  DS    F                 FPC after instruction
00001418  00000000 00000000                         970+         DS    FD                   gap
00001420  40404040 40404040                         971+         DC    CL8' '            was cross check skipped?
00001428  00000000 00000000                         972+         DS    FD                   gap
00001430  00000000 00000000                         973+XCO3     DS    XL16              Cross check Output
00001438  00000000 00000000
00001440  00000000 00000000                         974+         DS    XL16
00001448  00000000 00000000
00001450  00000000 00000000                         975+         DS    FD                   gap
00001458  00000000                                  976+FPC_XC_3 DS    F                1st cross check FPC
0000145C  00000000                                  977+         DS    F                2nd cross check FPC
00001460  00000000 00000000                         978+         DS    FD                   gap
00001468  00000000                                  979+         DS    F                 debug area
0000146C  00000000                                  980+         DS    F
                                                    981+*
00001470                                            982+X3       DS    0F
00001470  B29D 852C                      0000072C   983+         LFPC  FPCMOFF           initialize FPC
00001474  E320 500C 0014                 000013DC   984+         LGF   R2,V2_3           get v2
0000147A  E767 2000 0C36                 00000000   985+         VLM   V22,V23,0(R2)
00001480  E666 7002 0E75                            986+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
00001486  B29C 5040                      00001410   987+         STFPC FPC_R_3           save FPC
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

0000148A  E760 5028 080E                000013F8   988+         VST   V22,V1O3          save instruction result
00001490  07FB                                     989+         BR    R11               return
00001494                                           990+RE3      DS    0F                expected 16 byte result
00001494                                           991+         DROP  R5
00001494  3C000000 00000000                        992          DC    XL16'3C00000000000000 BC00000000000000'
0000149C  BC000000 00000000
000014A4  3F000000 00000000                        993          DC    XL16'3F00000000000000 0000000000000000'
000014AC  00000000 00000000
000014B4  BF000000 00000000                        994          DC    XL16'BF00000000000000 0000000000000000'
000014BC  00000000 00000000
                                                   995
                                                   996 * +1/64, -1/64
                                                   997          VRR_C VCRNF,0,2,00,00,N
000014C8                                           998+         DS    0FD
000014C8                      000014C8             999+         USING *,R5              base for test data and test routine
000014C8  00001568                                1000+T4       DC    A(X4)             address of test routine
000014CC  0004                                    1001+         DC    H'4'              test number
000014CE  00                                      1002+         DC    X'00'
000014CF  D5                                      1003+         DC    CL1'N'            Y = skip cross check
000014D0  00                                      1004+         DC    HL1'0'            m4
000014D1  02                                      1005+         DC    HL1'2'            m5
000014D2  00                                      1006+FLG4     DC    X'00'             expected FPC flags
000014D3  00                                      1007+VXC4     DC    X'00'             expected VXC
000014D4  0000159C                                1008+V2_4     DC    A(RE4+16)         address of v2: 16-byte packed decimal
000014D8  E5C3D9D5 C6404040                       1009+         DC    CL8'VCRNF'        instruction name
000014E0  00000010                                1010+         DC    A(16)             result length
000014E4  0000158C                                1011+         DC    A(RE4)            address of expected resul
000014E8  00000000 00000000                       1012+         DS    FD                gap
000014F0  00000000 00000000                       1013+V1O4     DS    XL16              V1 output
000014F8  00000000 00000000
00001500  00000000 00000000                       1014+         DS    FD                   gap
00001508  00000000                                1015+FPC_R_4  DS    F                 FPC after instruction
00001510  00000000 00000000                       1016+         DS    FD                   gap
00001518  40404040 40404040                       1017+         DC    CL8' '            was cross check skipped?
00001520  00000000 00000000                       1018+         DS    FD                   gap
00001528  00000000 00000000                       1019+XCO4     DS    XL16              Cross check Output
00001530  00000000 00000000
00001538  00000000 00000000                       1020+         DS    XL16
00001540  00000000 00000000
00001548  00000000 00000000                       1021+         DS    FD                   gap
00001550  00000000                                1022+FPC_XC_4 DS    F                1st cross check FPC
00001554  00000000                                1023+         DS    F                2nd cross check FPC
00001558  00000000 00000000                       1024+         DS    FD                   gap
00001560  00000000                                1025+         DS    F                 debug area
00001564  00000000                                1026+         DS    F
                                                  1027+*
00001568                                          1028+X4       DS    0F
00001568  B29D 852C                   0000072C    1029+         LFPC  FPCMOFF           initialize FPC
0000156C  E320 500C 0014              000014D4    1030+         LGF   R2,V2_4           get v2
00001572  E767 2000 0C36              00000000    1031+         VLM   V22,V23,0(R2)
00001578  E666 7002 0E75                          1032+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
0000157E  B29C 5040                   00001508    1033+         STFPC FPC_R_4           save FPC
00001582  E760 5028 080E              000014F0    1034+         VST   V22,V1O4          save instruction result
00001588  07FB                                    1035+         BR    R11               return
0000158C                                          1036+RE4      DS    0F                expected 16 byte result
0000158C                                          1037+         DROP  R5
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0000158C  32000000 00000000                        1038          DC    XL16'3200000000000000 B200000000000000'
00001594  B2000000 00000000
0000159C  3C800000 00000000                        1039          DC    XL16'3C80000000000000 0000000000000000'
000015A4  00000000 00000000
000015AC  BC800000 00000000                        1040          DC    XL16'BC80000000000000 0000000000000000'
000015B4  00000000 00000000
                                                   1041
                                                   1042 * +0, -0
                                                   1043          VRR_C VCRNF,0,2,00,00,N
000015C0                                           1044+         DS    0FD
000015C0                      000015C0             1045+         USING *,R5              base for test data and test routine
000015C0  00001660                                 1046+T5       DC    A(X5)             address of test routine
000015C4  0005                                     1047+         DC    H'5'              test number
000015C6  00                                       1048+         DC    X'00'
000015C7  D5                                       1049+         DC    CL1'N'            Y = skip cross check
000015C8  00                                       1050+         DC    HL1'0'            m4
000015C9  02                                       1051+         DC    HL1'2'            m5
000015CA  00                                       1052+FLG5     DC    X'00'             expected FPC flags
000015CB  00                                       1053+VXC5     DC    X'00'             expected VXC
000015CC  00001694                                 1054+V2_5     DC    A(RE5+16)         address of v2: 16-byte packed decimal
000015D0  E5C3D9D5 C6404040                        1055+         DC    CL8'VCRNF'        instruction name
000015D8  00000010                                 1056+         DC    A(16)             result length
000015DC  00001684                                 1057+         DC    A(RE5)            address of expected resul
000015E0  00000000 00000000                        1058+         DS    FD                gap
000015E8  00000000 00000000                        1059+V1O5     DS    XL16              V1 output
000015F0  00000000 00000000
000015F8  00000000 00000000                        1060+         DS    FD                   gap
00001600  00000000                                 1061+FPC_R_5  DS    F                 FPC after instruction
00001608  00000000 00000000                        1062+         DS    FD                   gap
00001610  40404040 40404040                        1063+         DC    CL8' '            was cross check skipped?
00001618  00000000 00000000                        1064+         DS    FD                   gap
00001620  00000000 00000000                        1065+XCO5     DS    XL16              Cross check Output
00001628  00000000 00000000
00001630  00000000 00000000                        1066+         DS    XL16
00001638  00000000 00000000
00001640  00000000 00000000                        1067+         DS    FD                   gap
00001648  00000000                                 1068+FPC_XC_5 DS    F                1st cross check FPC
0000164C  00000000                                 1069+         DS    F                2nd cross check FPC
00001650  00000000 00000000                        1070+         DS    FD                   gap
00001658  00000000                                 1071+         DS    F                 debug area
0000165C  00000000                                 1072+         DS    F
                                                   1073+*
00001660                                           1074+X5       DS    0F
00001660  B29D 852C                     0000072C   1075+         LFPC  FPCMOFF           initialize FPC
00001664  E320 500C 0014                000015CC   1076+         LGF   R2,V2_5           get v2
0000166A  E767 2000 0C36                00000000   1077+         VLM   V22,V23,0(R2)
00001670  E666 7002 0E75                           1078+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
00001676  B29C 5040                     00001600   1079+         STFPC FPC_R_5           save FPC
0000167A  E760 5028 080E                000015E8   1080+         VST   V22,V1O5          save instruction result
00001680  07FB                                     1081+         BR    R11               return
00001684                                           1082+RE5      DS    0F                expected 16 byte result
00001684                                           1083+         DROP  R5
00001684  00000000 00000000                        1084          DC    XL16'0000000000000000 8000000000000000'
0000168C  80000000 00000000
00001694  00000000 00000000                        1085          DC    XL16'0000000000000000 0000000000000000'
0000169C  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000016A4  80000000 00000000                        1086          DC    XL16'8000000000000000 0000000000000000'
000016AC  00000000 00000000
                                                   1087
                                                   1088 * +15, -15
                                                   1089          VRR_C VCRNF,0,2,00,00,N
000016B8                                           1090+         DS    0FD
000016B8                      000016B8             1091+         USING *,R5              base for test data and test routine
000016B8  00001758                                 1092+T6       DC    A(X6)             address of test routine
000016BC  0006                                     1093+         DC    H'6'              test number
000016BE  00                                       1094+         DC    X'00'
000016BF  D5                                       1095+         DC    CL1'N'            Y = skip cross check
000016C0  00                                       1096+         DC    HL1'0'            m4
000016C1  02                                       1097+         DC    HL1'2'            m5
000016C2  00                                       1098+FLG6     DC    X'00'             expected FPC flags
000016C3  00                                       1099+VXC6     DC    X'00'             expected VXC
000016C4  0000178C                                 1100+V2_6     DC    A(RE6+16)         address of v2: 16-byte packed decimal
000016C8  E5C3D9D5 C6404040                        1101+         DC    CL8'VCRNF'        instruction name
000016D0  00000010                                 1102+         DC    A(16)             result length
000016D4  0000177C                                 1103+         DC    A(RE6)            address of expected resul
000016D8  00000000 00000000                        1104+         DS    FD                gap
000016E0  00000000 00000000                        1105+V1O6     DS    XL16              V1 output
000016E8  00000000 00000000
000016F0  00000000 00000000                        1106+         DS    FD                   gap
000016F8  00000000                                 1107+FPC_R_6  DS    F                 FPC after instruction
00001700  00000000 00000000                        1108+         DS    FD                   gap
00001708  40404040 40404040                        1109+         DC    CL8' '            was cross check skipped?
00001710  00000000 00000000                        1110+         DS    FD                   gap
00001718  00000000 00000000                        1111+XCO6     DS    XL16              Cross check Output
00001720  00000000 00000000
00001728  00000000 00000000                        1112+         DS    XL16
00001730  00000000 00000000
00001738  00000000 00000000                        1113+         DS    FD                   gap
00001740  00000000                                 1114+FPC_XC_6 DS    F                1st cross check FPC
00001744  00000000                                 1115+         DS    F                2nd cross check FPC
00001748  00000000 00000000                        1116+         DS    FD                   gap
00001750  00000000                                 1117+         DS    F                 debug area
00001754  00000000                                 1118+         DS    F
                                                   1119+*
00001758                                           1120+X6       DS    0F
00001758  B29D 852C                     0000072C   1121+         LFPC  FPCMOFF           initialize FPC
0000175C  E320 500C 0014                000016C4   1122+         LGF   R2,V2_6           get v2
00001762  E767 2000 0C36                00000000   1123+         VLM   V22,V23,0(R2)
00001768  E666 7002 0E75                           1124+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
0000176E  B29C 5040                     000016F8   1125+         STFPC FPC_R_6           save FPC
00001772  E760 5028 080E                000016E0   1126+         VST   V22,V1O6          save instruction result
00001778  07FB                                     1127+         BR    R11               return
0000177C                                           1128+RE6      DS    0F                expected 16 byte result
0000177C                                           1129+         DROP  R5
0000177C  45C00000 00000000                        1130          DC    XL16'45C0000000000000 C5C0000000000000'
00001784  C5C00000 00000000
0000178C  41700000 00000000                        1131          DC    XL16'4170000000000000 0000000000000000'
00001794  00000000 00000000
0000179C  C1700000 00000000                        1132          DC    XL16'C170000000000000 0000000000000000'
000017A4  00000000 00000000
                                                   1133
                                                   1134 * +20/7, - 20/7
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   1135          VRR_C VCRNF,0,2,00,00,S                skip xc: lost digits
000017B0                                           1136+         DS    0FD
000017B0                     000017B0              1137+         USING *,R5              base for test data and test routine
000017B0  00001850                                 1138+T7       DC    A(X7)             address of test routine
000017B4  0007                                     1139+         DC    H'7'              test number
000017B6  00                                       1140+         DC    X'00'
000017B7  E2                                       1141+         DC    CL1'S'            Y = skip cross check
000017B8  00                                       1142+         DC    HL1'0'            m4
000017B9  02                                       1143+         DC    HL1'2'            m5
000017BA  00                                       1144+FLG7     DC    X'00'             expected FPC flags
000017BB  00                                       1145+VXC7     DC    X'00'             expected VXC
000017BC  00001884                                 1146+V2_7     DC    A(RE7+16)         address of v2: 16-byte packed decimal
000017C0  E5C3D9D5 C6404040                        1147+         DC    CL8'VCRNF'        instruction name
000017C8  00000010                                 1148+         DC    A(16)             result length
000017CC  00001874                                 1149+         DC    A(RE7)            address of expected resul
000017D0  00000000 00000000                        1150+         DS    FD                gap
000017D8  00000000 00000000                        1151+V1O7     DS    XL16              V1 output
000017E0  00000000 00000000
000017E8  00000000 00000000                        1152+         DS    FD                   gap
000017F0  00000000                                 1153+FPC_R_7  DS    F                 FPC after instruction
000017F8  00000000 00000000                        1154+         DS    FD                   gap
00001800  40404040 40404040                        1155+         DC    CL8' '            was cross check skipped?
00001808  00000000 00000000                        1156+         DS    FD                   gap
00001810  00000000 00000000                        1157+XCO7     DS    XL16              Cross check Output
00001818  00000000 00000000
00001820  00000000 00000000                        1158+         DS    XL16
00001828  00000000 00000000
00001830  00000000 00000000                        1159+         DS    FD                   gap
00001838  00000000                                 1160+FPC_XC_7 DS    F                1st cross check FPC
0000183C  00000000                                 1161+         DS    F                2nd cross check FPC
00001840  00000000 00000000                        1162+         DS    FD                   gap
00001848  00000000                                 1163+         DS    F                 debug area
0000184C  00000000                                 1164+         DS    F
                                                   1165+*
00001850                                           1166+X7       DS    0F
00001850  B29D 852C                     0000072C   1167+         LFPC  FPCMOFF           initialize FPC
00001854  E320 500C 0014                000017BC   1168+         LGF   R2,V2_7           get v2
0000185A  E767 2000 0C36                00000000   1169+         VLM   V22,V23,0(R2)
00001860  E666 7002 0E75                           1170+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
00001866  B29C 5040                     000017F0   1171+         STFPC FPC_R_7           save FPC
0000186A  E760 5028 080E                000017D8   1172+         VST   V22,V1O7          save instruction result
00001870  07FB                                     1173+         BR    R11               return
00001874                                           1174+RE7      DS    0F                expected 16 byte result
00001874                                           1175+         DROP  R5
00001874  40DB0000 00000000                        1176          DC    XL16'40DB000000000000 C0DB000000000000'
0000187C  C0DB0000 00000000
00001884  4036DB6E 00000000                        1177          DC    XL16'4036DB6E00000000 0000000000000000'
0000188C  00000000 00000000
00001894  C036DB6E 00000000                        1178          DC    XL16'C036DB6E00000000 0000000000000000'
0000189C  00000000 00000000
                                                   1179
                                                   1180 * +2^(-126), -2^(-126)
                                                   1181          VRR_C VCRNF,0,2,10,00,S                skip xc: underflow
000018A8                                           1182+         DS    0FD
000018A8                     000018A8              1183+         USING *,R5              base for test data and test routine
000018A8  00001948                                 1184+T8       DC    A(X8)             address of test routine
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000018AC  0008                                     1185+         DC    H'8'              test number
000018AE  00                                       1186+         DC    X'00'
000018AF  E2                                       1187+         DC    CL1'S'            Y = skip cross check
000018B0  00                                       1188+         DC    HL1'0'            m4
000018B1  02                                       1189+         DC    HL1'2'            m5
000018B2  10                                       1190+FLG8     DC    X'10'             expected FPC flags
000018B3  00                                       1191+VXC8     DC    X'00'             expected VXC
000018B4  0000197C                                 1192+V2_8     DC    A(RE8+16)         address of v2: 16-byte packed decimal
000018B8  E5C3D9D5 C6404040                        1193+         DC    CL8'VCRNF'        instruction name
000018C0  00000010                                 1194+         DC    A(16)             result length
000018C4  0000196C                                 1195+         DC    A(RE8)            address of expected resul
000018C8  00000000 00000000                        1196+         DS    FD                gap
000018D0  00000000 00000000                        1197+V1O8     DS    XL16              V1 output
000018D8  00000000 00000000
000018E0  00000000 00000000                        1198+         DS    FD                   gap
000018E8  00000000                                 1199+FPC_R_8  DS    F                 FPC after instruction
000018F0  00000000 00000000                        1200+         DS    FD                   gap
000018F8  40404040 40404040                        1201+         DC    CL8' '            was cross check skipped?
00001900  00000000 00000000                        1202+         DS    FD                   gap
00001908  00000000 00000000                        1203+XCO8     DS    XL16              Cross check Output
00001910  00000000 00000000
00001918  00000000 00000000                        1204+         DS    XL16
00001920  00000000 00000000
00001928  00000000 00000000                        1205+         DS    FD                   gap
00001930  00000000                                 1206+FPC_XC_8 DS    F                1st cross check FPC
00001934  00000000                                 1207+         DS    F                2nd cross check FPC
00001938  00000000 00000000                        1208+         DS    FD                   gap
00001940  00000000                                 1209+         DS    F                 debug area
00001944  00000000                                 1210+         DS    F
                                                   1211+*
00001948                                           1212+X8       DS    0F
00001948  B29D 852C                      0000072C  1213+         LFPC  FPCMOFF           initialize FPC
0000194C  E320 500C 0014                 000018B4  1214+         LGF   R2,V2_8           get v2
00001952  E767 2000 0C36                 00000000  1215+         VLM   V22,V23,0(R2)
00001958  E666 7002 0E75                           1216+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
0000195E  B29C 5040                      000018E8  1217+         STFPC FPC_R_8           save FPC
00001962  E760 5028 080E                 000018D0  1218+         VST   V22,V1O8          save instruction result
00001968  07FB                                     1219+         BR    R11               return
0000196C                                           1220+RE8      DS    0F                expected 16 byte result
0000196C                                           1221+         DROP  R5
0000196C  00000000 00000000                        1222          DC    XL16'0000000000000000 8000000000000000'
00001974  80000000 00000000
0000197C  00800000 00000000                        1223          DC    XL16'0080000000000000 0000000000000000'
00001984  00000000 00000000
0000198C  80800000 00000000                        1224          DC    XL16'8080000000000000 0000000000000000'
00001994  00000000 00000000
                                                   1225
                                                   1226 * +2^(-149), -2^(-149) - subnormal
                                                   1227          VRR_C VCRNF,0,2,10,00,S                 skip xc: underflow
000019A0                                           1228+         DS    0FD
000019A0                         000019A0          1229+         USING *,R5              base for test data and test routine
000019A0  00001A40                                 1230+T9       DC    A(X9)             address of test routine
000019A4  0009                                     1231+         DC    H'9'              test number
000019A6  00                                       1232+         DC    X'00'
000019A7  E2                                       1233+         DC    CL1'S'            Y = skip cross check
000019A8  00                                       1234+         DC    HL1'0'            m4
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

000019A9  02                                       1235+          DC    HL1'2'            m5
000019AA  10                                       1236+FLG9      DC    X'10'             expected FPC flags
000019AB  00                                       1237+VXC9      DC    X'00'             expected VXC
000019AC  00001A74                                 1238+V2_9      DC    A(RE9+16)         address of v2: 16-byte packed decimal
000019B0  E5C3D9D5 C6404040                        1239+          DC    CL8'VCRNF'        instruction name
000019B8  00000010                                 1240+          DC    A(16)             result length
000019BC  00001A64                                 1241+          DC    A(RE9)            address of expected resul
000019C0  00000000 00000000                        1242+          DS    FD                gap
000019C8  00000000 00000000                        1243+V1O9      DS    XL16              V1 output
000019D0  00000000 00000000
000019D8  00000000 00000000                        1244+          DS    FD                   gap
000019E0  00000000                                 1245+FPC_R_9   DS    F                 FPC after instruction
000019E8  00000000 00000000                        1246+          DS    FD                   gap
000019F0  40404040 40404040                        1247+          DC    CL8' '            was cross check skipped?
000019F8  00000000 00000000                        1248+          DS    FD                   gap
00001A00  00000000 00000000                        1249+XCO9      DS    XL16              Cross check Output
00001A08  00000000 00000000
00001A10  00000000 00000000                        1250+          DS    XL16
00001A18  00000000 00000000
00001A20  00000000 00000000                        1251+          DS    FD                   gap
00001A28  00000000                                 1252+FPC_XC_9  DS    F                1st cross check FPC
00001A2C  00000000                                 1253+          DS    F                2nd cross check FPC
00001A30  00000000 00000000                        1254+          DS    FD                   gap
00001A38  00000000                                 1255+          DS    F                 debug area
00001A3C  00000000                                 1256+          DS    F
                                                   1257+*
00001A40                                           1258+X9        DS    0F
00001A40  B29D 852C                     0000072C   1259+          LFPC  FPCMOFF           initialize FPC
00001A44  E320 500C 0014                000019AC   1260+          LGF   R2,V2_9           get v2
00001A4A  E767 2000 0C36                00000000   1261+          VLM   V22,V23,0(R2)
00001A50  E666 7002 0E75                           1262+          VCRNF V22,V22,V23,0,2       test instruction (dest is source)
00001A56  B29C 5040                     000019E0   1263+          STFPC FPC_R_9           save FPC
00001A5A  E760 5028 080E                000019C8   1264+          VST   V22,V1O9          save instruction result
00001A60  07FB                                     1265+          BR    R11               return
00001A64                                           1266+RE9       DS    0F                expected 16 byte result
00001A64                                           1267+          DROP  R5
00001A64  00000000 00000000                        1268           DC    XL16'0000000000000000 8000000000000000'
00001A6C  80000000 00000000
00001A74  00000001 00000000                        1269           DC    XL16'0000000100000000 0000000000000000'
00001A7C  00000000 00000000
00001A84  80800001 00000000                        1270           DC    XL16'8080000100000000 0000000000000000'
00001A8C  00000000 00000000
                                                   1271
                                                   1272 * +2^(128) * (1 - 2^(-24)) , - +2^(128) * (1 - 2^(-24)) : 3402823ï€¥10+32
                                                   1273           VRR_C VCRNF,0,2,20,00,S                 skip xc: overflow
00001A98                                           1274+          DS    0FD
00001A98                        00001A98           1275+          USING *,R5              base for test data and test routine
00001A98  00001B38                                 1276+T10       DC    A(X10)            address of test routine
00001A9C  000A                                     1277+          DC    H'10'             test number
00001A9E  00                                       1278+          DC    X'00'
00001A9F  E2                                       1279+          DC    CL1'S'            Y = skip cross check
00001AA0  00                                       1280+          DC    HL1'0'            m4
00001AA1  02                                       1281+          DC    HL1'2'            m5
00001AA2  20                                       1282+FLG10     DC    X'20'             expected FPC flags
00001AA3  00                                       1283+VXC10     DC    X'00'             expected VXC
00001AA4  00001B6C                                 1284+V2_10     DC    A(RE10+16)        address of v2: 16-byte packed decimal
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001AA8  E5C3D9D5 C6404040                        1285+         DC    CL8'VCRNF'        instruction name
00001AB0  00000010                                 1286+         DC    A(16)             result length
00001AB4  00001B5C                                 1287+         DC    A(RE10)           address of expected resul
00001AB8  00000000 00000000                        1288+         DS    FD                gap
00001AC0  00000000 00000000                        1289+V1O10    DS    XL16              V1 output
00001AC8  00000000 00000000
00001AD0  00000000 00000000                        1290+         DS    FD                   gap
00001AD8  00000000                                 1291+FPC_R_10 DS    F                 FPC after instruction
00001AE0  00000000 00000000                        1292+         DS    FD                   gap
00001AE8  40404040 40404040                        1293+         DC    CL8' '            was cross check skipped?
00001AF0  00000000 00000000                        1294+         DS    FD                   gap
00001AF8  00000000 00000000                        1295+XCO10    DS    XL16              Cross check Output
00001B00  00000000 00000000
00001B08  00000000 00000000                        1296+         DS    XL16
00001B10  00000000 00000000
00001B18  00000000 00000000                        1297+         DS    FD                   gap
00001B20  00000000                                 1298+FPC_XC_10 DS   F                1st cross check FPC
00001B24  00000000                                 1299+         DS    F                2nd cross check FPC
00001B28  00000000 00000000                        1300+         DS    FD                   gap
00001B30  00000000                                 1301+         DS    F                 debug area
00001B34  00000000                                 1302+         DS    F
                                                   1303+*
00001B38                                           1304+X10      DS    0F
00001B38  B29D 852C                     0000072C   1305+         LFPC  FPCMOFF           initialize FPC
00001B3C  E320 500C 0014                00001AA4   1306+         LGF   R2,V2_10          get v2
00001B42  E767 2000 0C36                00000000   1307+         VLM   V22,V23,0(R2)
00001B48  E666 7002 0E75                           1308+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
00001B4E  B29C 5040                     00001AD8   1309+         STFPC FPC_R_10          save FPC
00001B52  E760 5028 080E                00001AC0   1310+         VST   V22,V1O10         save instruction result
00001B58  07FB                                     1311+         BR    R11               return
00001B5C                                           1312+RE10     DS    0F                expected 16 byte result
00001B5C                                           1313+         DROP  R5
00001B5C  7FFE0000 00000000                        1314          DC    XL16'7FFE000000000000 FFFE000000000000'
00001B64  FFFE0000 00000000
00001B6C  7F7FFFFF 00000000                        1315          DC    XL16'7F7FFFFF00000000 0000000000000000'
00001B74  00000000 00000000
00001B7C  FF7FFFFF 00000000                        1316          DC    XL16'FF7FFFFF00000000 0000000000000000'
00001B84  00000000 00000000
                                                   1317
                                                   1318 * NAN, -NAN
                                                   1319          VRR_C VCRNF,0,2,80,00,S                 skip xc: invalid on XC
00001B90                                           1320+         DS    0FD
00001B90                        00001B90           1321+         USING *,R5              base for test data and test routine
00001B90  00001C30                                 1322+T11      DC    A(X11)            address of test routine
00001B94  000B                                     1323+         DC    H'11'             test number
00001B96  00                                       1324+         DC    X'00'
00001B97  E2                                       1325+         DC    CL1'S'            Y = skip cross check
00001B98  00                                       1326+         DC    HL1'0'            m4
00001B99  02                                       1327+         DC    HL1'2'            m5
00001B9A  80                                       1328+FLG11    DC    X'80'             expected FPC flags
00001B9B  00                                       1329+VXC11    DC    X'00'             expected VXC
00001B9C  00001C64                                 1330+V2_11    DC    A(RE11+16)        address of v2: 16-byte packed decimal
00001BA0  E5C3D9D5 C6404040                        1331+         DC    CL8'VCRNF'        instruction name
00001BA8  00000010                                 1332+         DC    A(16)             result length
00001BAC  00001C54                                 1333+         DC    A(RE11)           address of expected resul
00001BB0  00000000 00000000                        1334+         DS    FD                gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001BB8  00000000 00000000                        1335+V1O11    DS    XL16              V1 output
00001BC0  00000000 00000000
00001BC8  00000000 00000000                        1336+         DS    FD                   gap
00001BD0  00000000                                 1337+FPC_R_11 DS    F                 FPC after instruction
00001BD8  00000000 00000000                        1338+         DS    FD                   gap
00001BE0  40404040 40404040                        1339+         DC    CL8' '            was cross check skipped?
00001BE8  00000000 00000000                        1340+         DS    FD                   gap
00001BF0  00000000 00000000                        1341+XCO11    DS    XL16              Cross check Output
00001BF8  00000000 00000000
00001C00  00000000 00000000                        1342+         DS    XL16
00001C08  00000000 00000000
00001C10  00000000 00000000                        1343+         DS    FD                   gap
00001C18  00000000                                 1344+FPC_XC_11 DS   F                1st cross check FPC
00001C1C  00000000                                 1345+         DS    F                2nd cross check FPC
00001C20  00000000 00000000                        1346+         DS    FD                   gap
00001C28  00000000                                 1347+         DS    F                 debug area
00001C2C  00000000                                 1348+         DS    F
                                                   1349+*
00001C30                                           1350+X11      DS    0F
00001C30  B29D 852C                     0000072C   1351+         LFPC  FPCMOFF           initialize FPC
00001C34  E320 500C 0014                00001B9C   1352+         LGF   R2,V2_11          get v2
00001C3A  E767 2000 0C36                00000000   1353+         VLM   V22,V23,0(R2)
00001C40  E666 7002 0E75                           1354+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
00001C46  B29C 5040                     00001BD0   1355+         STFPC FPC_R_11          save FPC
00001C4A  E760 5028 080E                00001BB8   1356+         VST   V22,V1O11         save instruction result
00001C50  07FB                                     1357+         BR    R11               return
00001C54                                           1358+RE11     DS    0F                expected 16 byte result
00001C54                                           1359+         DROP  R5
00001C54  7FFF0000 00000000                        1360          DC    XL16'7FFF000000000000 FFFF000000000000'
00001C5C  FFFF0000 00000000
00001C64  7FC00000 00000000                        1361          DC    XL16'7FC0000000000000 0000000000000000'
00001C6C  00000000 00000000
00001C74  FFC00000 00000000                        1362          DC    XL16'FFC0000000000000 0000000000000000'
00001C7C  00000000 00000000
                                                   1363
                                                   1364 * bad m4 - inexact
                                                   1365          VRR_C VCRNF,2,2,08,00,S                skip xc: inexact
00001C88                                           1366+         DS    0FD
00001C88                        00001C88           1367+         USING *,R5              base for test data and test routine
00001C88  00001D28                                 1368+T12      DC    A(X12)            address of test routine
00001C8C  000C                                     1369+         DC    H'12'             test number
00001C8E  00                                       1370+         DC    X'00'
00001C8F  E2                                       1371+         DC    CL1'S'            Y = skip cross check
00001C90  02                                       1372+         DC    HL1'2'            m4
00001C91  02                                       1373+         DC    HL1'2'            m5
00001C92  08                                       1374+FLG12    DC    X'08'             expected FPC flags
00001C93  00                                       1375+VXC12    DC    X'00'             expected VXC
00001C94  00001D5C                                 1376+V2_12    DC    A(RE12+16)        address of v2: 16-byte packed decimal
00001C98  E5C3D9D5 C6404040                        1377+         DC    CL8'VCRNF'        instruction name
00001CA0  00000010                                 1378+         DC    A(16)             result length
00001CA4  00001D4C                                 1379+         DC    A(RE12)           address of expected resul
00001CA8  00000000 00000000                        1380+         DS    FD                gap
00001CB0  00000000 00000000                        1381+V1O12    DS    XL16              V1 output
00001CB8  00000000 00000000
00001CC0  00000000 00000000                        1382+         DS    FD                   gap
00001CC8  00000000                                 1383+FPC_R_12 DS    F                 FPC after instruction
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001CD0  00000000 00000000                        1384+          DS    FD                    gap
00001CD8  40404040 40404040                        1385+          DC    CL8' '             was cross check skipped?
00001CE0  00000000 00000000                        1386+          DS    FD                    gap
00001CE8  00000000 00000000                        1387+XCO12     DS    XL16               Cross check Output
00001CF0  00000000 00000000
00001CF8  00000000 00000000                        1388+          DS    XL16
00001D00  00000000 00000000
00001D08  00000000 00000000                        1389+          DS    FD                    gap
00001D10  00000000                                 1390+FPC_XC_12 DS    F                  1st cross check FPC
00001D14  00000000                                 1391+          DS    F                  2nd cross check FPC
00001D18  00000000 00000000                        1392+          DS    FD                    gap
00001D20  00000000                                 1393+          DS    F                  debug area
00001D24  00000000                                 1394+          DS    F
                                                   1395+*
00001D28                                           1396+X12       DS    0F
00001D28  B29D 852C                     0000072C   1397+          LFPC  FPCMOFF            initialize FPC
00001D2C  E320 500C 0014                00001C94   1398+          LGF   R2,V2_12           get v2
00001D32  E767 2000 0C36                00000000   1399+          VLM   V22,V23,0(R2)
00001D38  E666 7002 2E75                           1400+          VCRNF V22,V22,V23,2,2       test instruction (dest is source)
00001D3E  B29C 5040                     00001CC8   1401+          STFPC FPC_R_12           save FPC
00001D42  E760 5028 080E                00001CB0   1402+          VST   V22,V1O12          save instruction result
00001D48  07FB                                     1403+          BR    R11                return
00001D4C                                           1404+RE12      DS    0F                 expected 16 byte result
00001D4C                                           1405+          DROP  R5
00001D4C  00000000 00000000                        1406           DC    XL16'0000000000000000 0000000000000000'
00001D54  00000000 00000000
00001D5C  7FC00000 00000000                        1407           DC    XL16'7FC0000000000000 0000000000000000'
00001D64  00000000 00000000
00001D6C  FFC00000 00000000                        1408           DC    XL16'FFC0000000000000 0000000000000000'
00001D74  00000000 00000000
                                                   1409
                                                   1410 * bad m5 - inexact
                                                   1411           VRR_C VCRNF,0,3,08,00,S                 skip xc: inexact
00001D80                                           1412+          DS    0FD
00001D80                        00001D80           1413+          USING *,R5               base for test data and test routine
00001D80  00001E20                                 1414+T13       DC    A(X13)             address of test routine
00001D84  000D                                     1415+          DC    H'13'              test number
00001D86  00                                       1416+          DC    X'00'
00001D87  E2                                       1417+          DC    CL1'S'             Y = skip cross check
00001D88  00                                       1418+          DC    HL1'0'             m4
00001D89  03                                       1419+          DC    HL1'3'             m5
00001D8A  08                                       1420+FLG13     DC    X'08'              expected FPC flags
00001D8B  00                                       1421+VXC13     DC    X'00'              expected VXC
00001D8C  00001E54                                 1422+V2_13     DC    A(RE13+16)         address of v2: 16-byte packed decimal
00001D90  E5C3D9D5 C6404040                        1423+          DC    CL8'VCRNF'         instruction name
00001D98  00000010                                 1424+          DC    A(16)              result length
00001D9C  00001E44                                 1425+          DC    A(RE13)            address of expected resul
00001DA0  00000000 00000000                        1426+          DS    FD                 gap
00001DA8  00000000 00000000                        1427+V1O13     DS    XL16               V1 output
00001DB0  00000000 00000000
00001DB8  00000000 00000000                        1428+          DS    FD                    gap
00001DC0  00000000                                 1429+FPC_R_13  DS    F                  FPC after instruction
00001DC8  00000000 00000000                        1430+          DS    FD                    gap
00001DD0  40404040 40404040                        1431+          DC    CL8' '             was cross check skipped?
00001DD8  00000000 00000000                        1432+          DS    FD                    gap
00001DE0  00000000 00000000                        1433+XCO13     DS    XL16               Cross check Output
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00001DE8  00000000 00000000
00001DF0  00000000 00000000                         1434+         DS    XL16
00001DF8  00000000 00000000
00001E00  00000000 00000000                         1435+         DS    FD                    gap
00001E08  00000000                                  1436+FPC_XC_13 DS   F                 1st cross check FPC
00001E0C  00000000                                  1437+         DS    F                 2nd cross check FPC
00001E10  00000000 00000000                         1438+         DS    FD                    gap
00001E18  00000000                                  1439+         DS    F                  debug area
00001E1C  00000000                                  1440+         DS    F
                                                    1441+*
00001E20                                            1442+X13      DS    0F
00001E20  B29D 852C                     0000072C    1443+         LFPC  FPCMOFF            initialize FPC
00001E24  E320 500C 0014                00001D8C    1444+         LGF   R2,V2_13           get v2
00001E2A  E767 2000 0C36                00000000    1445+         VLM   V22,V23,0(R2)
00001E30  E666 7003 0E75                            1446+         VCRNF V22,V22,V23,0,3       test instruction (dest is source)
00001E36  B29C 5040                     00001DC0    1447+         STFPC FPC_R_13           save FPC
00001E3A  E760 5028 080E                00001DA8    1448+         VST   V22,V1O13          save instruction result
00001E40  07FB                                      1449+         BR    R11                return
00001E44                                            1450+RE13     DS    0F                 expected 16 byte result
00001E44                                            1451+         DROP  R5
00001E44  00000000 00000000                         1452          DC    XL16'0000000000000000 0000000000000000'
00001E4C  00000000 00000000
00001E54  7FC00000 00000000                         1453          DC    XL16'7FC0000000000000 0000000000000000'
00001E5C  00000000 00000000
00001E64  FFC00000 00000000                         1454          DC    XL16'FFC0000000000000 0000000000000000'
00001E6C  00000000 00000000
                                                    1455
                                                    1456 * Additional tests
                                                    1457 *-----------------------------------------------------------------
                                                    1458 * +99.1, -99.1
                                                    1459          VRR_C VCRNF,0,2,00,00,S                skip xc: lost digits
00001E78                                            1460+         DS    0FD
00001E78                        00001E78            1461+         USING *,R5             base for test data and test routine
00001E78  00001F18                                  1462+T14      DC    A(X14)           address of test routine
00001E7C  000E                                      1463+         DC    H'14'            test number
00001E7E  00                                        1464+         DC    X'00'
00001E7F  E2                                        1465+         DC    CL1'S'           Y = skip cross check
00001E80  00                                        1466+         DC    HL1'0'           m4
00001E81  02                                        1467+         DC    HL1'2'           m5
00001E82  00                                        1468+FLG14    DC    X'00'            expected FPC flags
00001E83  00                                        1469+VXC14    DC    X'00'            expected VXC
00001E84  00001F4C                                  1470+V2_14    DC    A(RE14+16)       address of v2: 16-byte packed decimal
00001E88  E5C3D9D5 C6404040                         1471+         DC    CL8'VCRNF'       instruction name
00001E90  00000010                                  1472+         DC    A(16)            result length
00001E94  00001F3C                                  1473+         DC    A(RE14)          address of expected resul
00001E98  00000000 00000000                         1474+         DS    FD               gap
00001EA0  00000000 00000000                         1475+V1O14    DS    XL16             V1 output
00001EA8  00000000 00000000
00001EB0  00000000 00000000                         1476+         DS    FD                    gap
00001EB8  00000000                                  1477+FPC_R_14 DS    F                FPC after instruction
00001EC0  00000000 00000000                         1478+         DS    FD                    gap
00001EC8  40404040 40404040                         1479+         DC    CL8' '           was cross check skipped?
00001ED0  00000000 00000000                         1480+         DS    FD                    gap
00001ED8  00000000 00000000                         1481+XCO14    DS    XL16             Cross check Output
00001EE0  00000000 00000000
00001EE8  00000000 00000000                         1482+         DS    XL16
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00001EF0  00000000 00000000
00001EF8  00000000 00000000                        1483+         DS    FD                   gap
00001F00  00000000                                 1484+FPC_XC_14 DS   F                 1st cross check FPC
00001F04  00000000                                 1485+         DS    F                 2nd cross check FPC
00001F08  00000000 00000000                        1486+         DS    FD                   gap
00001F10  00000000                                 1487+         DS    F                  debug area
00001F14  00000000                                 1488+         DS    F
                                                   1489+*
00001F18                                           1490+X14      DS    0F
00001F18  B29D 852C                    0000072C    1491+         LFPC  FPCMOFF            initialize FPC
00001F1C  E320 500C 0014               00001E84    1492+         LGF   R2,V2_14           get v2
00001F22  E767 2000 0C36               00000000    1493+         VLM   V22,V23,0(R2)
00001F28  E666 7002 0E75                           1494+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
00001F2E  B29C 5040                    00001EB8    1495+         STFPC FPC_R_14           save FPC
00001F32  E760 5028 080E               00001EA0    1496+         VST   V22,V1O14          save instruction result
00001F38  07FB                                     1497+         BR    R11                return
00001F3C                                           1498+RE14     DS    0F                 expected 16 byte result
00001F3C                                           1499+         DROP  R5
00001F3C  4B190000 00000000                        1500          DC    XL16'4B19000000000000 CB19000000000000'
00001F44  CB190000 00000000
00001F4C  42C63333 00000000                        1501          DC    XL16'42C6333300000000 0000000000000000'
00001F54  00000000 00000000
00001F5C  C2C63333 00000000                        1502          DC    XL16'C2C6333300000000 0000000000000000'
00001F64  00000000 00000000
                                                   1503
                                                   1504 * +100.9, -100.9
                                                   1505          VRR_C VCRNF,0,2,00,00,S                 skip xc: lost digits
00001F70                                           1506+         DS    0FD
00001F70                       00001F70            1507+         USING *,R5               base for test data and test routine
00001F70  00002010                                 1508+T15      DC    A(X15)             address of test routine
00001F74  000F                                     1509+         DC    H'15'              test number
00001F76  00                                       1510+         DC    X'00'
00001F77  E2                                       1511+         DC    CL1'S'             Y = skip cross check
00001F78  00                                       1512+         DC    HL1'0'             m4
00001F79  02                                       1513+         DC    HL1'2'             m5
00001F7A  00                                       1514+FLG15    DC    X'00'              expected FPC flags
00001F7B  00                                       1515+VXC15    DC    X'00'              expected VXC
00001F7C  00002044                                 1516+V2_15    DC    A(RE15+16)         address of v2: 16-byte packed decimal
00001F80  E5C3D9D5 C6404040                        1517+         DC    CL8'VCRNF'         instruction name
00001F88  00000010                                 1518+         DC    A(16)              result length
00001F8C  00002034                                 1519+         DC    A(RE15)            address of expected resul
00001F90  00000000 00000000                        1520+         DS    FD                 gap
00001F98  00000000 00000000                        1521+V1O15    DS    XL16               V1 output
00001FA0  00000000 00000000
00001FA8  00000000 00000000                        1522+         DS    FD                   gap
00001FB0  00000000                                 1523+FPC_R_15 DS    F                  FPC after instruction
00001FB8  00000000 00000000                        1524+         DS    FD                   gap
00001FC0  40404040 40404040                        1525+         DC    CL8' '             was cross check skipped?
00001FC8  00000000 00000000                        1526+         DS    FD                   gap
00001FD0  00000000 00000000                        1527+XCO15    DS    XL16               Cross check Output
00001FD8  00000000 00000000
00001FE0  00000000 00000000                        1528+         DS    XL16
00001FE8  00000000 00000000
00001FF0  00000000 00000000                        1529+         DS    FD                   gap
00001FF8  00000000                                 1530+FPC_XC_15 DS   F                 1st cross check FPC
00001FFC  00000000                                 1531+         DS    F                 2nd cross check FPC
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00002000  00000000 00000000                       1532+         DS    FD                   gap
00002008  00000000                                1533+         DS    F                  debug area
0000200C  00000000                                1534+         DS    F
                                                  1535+*
00002010                                          1536+X15      DS    0F
00002010  B29D 852C                     0000072C  1537+         LFPC  FPCMOFF            initialize FPC
00002014  E320 500C 0014                00001F7C  1538+         LGF   R2,V2_15           get v2
0000201A  E767 2000 0C36                00000000  1539+         VLM   V22,V23,0(R2)
00002020  E666 7002 0E75                          1540+         VCRNF V22,V22,V23,0,2        test instruction (dest is source)
00002026  B29C 5040                     00001FB0  1541+         STFPC FPC_R_15           save FPC
0000202A  E760 5028 080E                00001F98  1542+         VST   V22,V1O15          save instruction result
00002030  07FB                                    1543+         BR    R11                return
00002034                                          1544+RE15     DS    0F                 expected 16 byte result
00002034                                          1545+         DROP  R5
00002034  4B270000 00000000                       1546          DC    XL16'4B27000000000000 CB27000000000000'
0000203C  CB270000 00000000
00002044  42C9CCCD 00000000                       1547          DC    XL16'42C9CCCD00000000 0000000000000000'
0000204C  00000000 00000000
00002054  C2C9CCCD 00000000                       1548          DC    XL16'C2C9CCCD00000000 0000000000000000'
0000205C  00000000 00000000
                                                  1549
                                                  1550 * +527.7895, -527.7895
                                                  1551          VRR_C VCRNF,0,2,00,00,S                skip xc: lost digits
00002068                                          1552+         DS    0FD
00002068                        00002068          1553+         USING *,R5               base for test data and test routine
00002068  00002108                                1554+T16      DC    A(X16)             address of test routine
0000206C  0010                                    1555+         DC    H'16'              test number
0000206E  00                                      1556+         DC    X'00'
0000206F  E2                                      1557+         DC    CL1'S'             Y = skip cross check
00002070  00                                      1558+         DC    HL1'0'             m4
00002071  02                                      1559+         DC    HL1'2'             m5
00002072  00                                      1560+FLG16    DC    X'00'              expected FPC flags
00002073  00                                      1561+VXC16    DC    X'00'              expected VXC
00002074  0000213C                                1562+V2_16    DC    A(RE16+16)         address of v2: 16-byte packed decimal
00002078  E5C3D9D5 C6404040                       1563+         DC    CL8'VCRNF'         instruction name
00002080  00000010                                1564+         DC    A(16)              result length
00002084  0000212C                                1565+         DC    A(RE16)            address of expected resul
00002088  00000000 00000000                       1566+         DS    FD                 gap
00002090  00000000 00000000                       1567+V1O16    DS    XL16               V1 output
00002098  00000000 00000000
000020A0  00000000 00000000                       1568+         DS    FD                   gap
000020A8  00000000                                1569+FPC_R_16 DS    F                  FPC after instruction
000020B0  00000000 00000000                       1570+         DS    FD                   gap
000020B8  40404040 40404040                       1571+         DC    CL8' '             was cross check skipped?
000020C0  00000000 00000000                       1572+         DS    FD                   gap
000020C8  00000000 00000000                       1573+XCO16    DS    XL16               Cross check Output
000020D0  00000000 00000000
000020D8  00000000 00000000                       1574+         DS    XL16
000020E0  00000000 00000000
000020E8  00000000 00000000                       1575+         DS    FD                   gap
000020F0  00000000                                1576+FPC_XC_16 DS   F                1st cross check FPC
000020F4  00000000                                1577+         DS    F                2nd cross check FPC
000020F8  00000000 00000000                       1578+         DS    FD                   gap
00002100  00000000                                1579+         DS    F                  debug area
00002104  00000000                                1580+         DS    F
                                                  1581+*
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00002108                                          1582+X16      DS    0F
00002108  B29D 852C                     0000072C  1583+         LFPC  FPCMOFF           initialize FPC
0000210C  E320 500C 0014                00002074  1584+         LGF   R2,V2_16          get v2
00002112  E767 2000 0C36                00000000  1585+         VLM   V22,V23,0(R2)
00002118  E666 7002 0E75                          1586+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
0000211E  B29C 5040                     000020A8  1587+         STFPC FPC_R_16          save FPC
00002122  E760 5028 080E                00002090  1588+         VST   V22,V1O16         save instruction result
00002128  07FB                                    1589+         BR    R11               return
0000212C                                          1590+RE16     DS    0F                expected 16 byte result
0000212C                                          1591+         DROP  R5
0000212C  50100000 00000000                       1592          DC    XL16'501000000000000 D010000000000000'
00002134  D0100000 00000000
0000213C  4403F287 00000000                       1593          DC    XL16'4403F28700000000 0000000000000000'
00002144  00000000 00000000
0000214C  C403F287 00000000                       1594          DC    XL16'C403F28700000000 0000000000000000'
00002154  00000000 00000000
                                                  1595
                                                  1596 * +1055.5790, -1055.5790
                                                  1597          VRR_C VCRNF,0,2,00,00,S                 skip xc: lost digits
00002160                                          1598+         DS    0FD
00002160                        00002160          1599+         USING *,R5              base for test data and test routine
00002160  00002200                                1600+T17      DC    A(X17)            address of test routine
00002164  0011                                    1601+         DC    H'17'             test number
00002166  00                                      1602+         DC    X'00'
00002167  E2                                      1603+         DC    CL1'S'            Y = skip cross check
00002168  00                                      1604+         DC    HL1'0'            m4
00002169  02                                      1605+         DC    HL1'2'            m5
0000216A  00                                      1606+FLG17    DC    X'00'             expected FPC flags
0000216B  00                                      1607+VXC17    DC    X'00'             expected VXC
0000216C  00002234                                1608+V2_17    DC    A(RE17+16)        address of v2: 16-byte packed decimal
00002170  E5C3D9D5 C6404040                       1609+         DC    CL8'VCRNF'        instruction name
00002178  00000010                                1610+         DC    A(16)             result length
0000217C  00002224                                1611+         DC    A(RE17)           address of expected resul
00002180  00000000 00000000                       1612+         DS    FD                gap
00002188  00000000 00000000                       1613+V1O17    DS    XL16              V1 output
00002190  00000000 00000000
00002198  00000000 00000000                       1614+         DS    FD                   gap
000021A0  00000000                                1615+FPC_R_17 DS    F                 FPC after instruction
000021A8  00000000 00000000                       1616+         DS    FD                   gap
000021B0  40404040 40404040                       1617+         DC    CL8' '            was cross check skipped?
000021B8  00000000 00000000                       1618+         DS    FD                   gap
000021C0  00000000 00000000                       1619+XCO17    DS    XL16              Cross check Output
000021C8  00000000 00000000
000021D0  00000000 00000000                       1620+         DS    XL16
000021D8  00000000 00000000
000021E0  00000000 00000000                       1621+         DS    FD                   gap
000021E8  00000000                                1622+FPC_XC_17 DS   F                1st cross check FPC
000021EC  00000000                                1623+         DS    F                2nd cross check FPC
000021F0  00000000 00000000                       1624+         DS    FD                   gap
000021F8  00000000                                1625+         DS    F                 debug area
000021FC  00000000                                1626+         DS    F
                                                  1627+*
00002200                                          1628+X17      DS    0F
00002200  B29D 852C                     0000072C  1629+         LFPC  FPCMOFF           initialize FPC
00002204  E320 500C 0014                0000216C  1630+         LGF   R2,V2_17          get v2
0000220A  E767 2000 0C36                00000000  1631+         VLM   V22,V23,0(R2)
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

00002210  E666 7002 0E75                          1632+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
00002216  B29C 5040                     000021A0  1633+         STFPC FPC_R_17          save FPC
0000221A  E760 5028 080E                00002188  1634+         VST   V22,V1O17         save instruction result
00002220  07FB                                    1635+         BR    R11               return
00002224                                          1636+RE17     DS    0F                expected 16 byte result
00002224                                          1637+         DROP  R5
00002224  52100000 00000000                       1638          DC    XL16'521000000000000 D210000000000000'
0000222C  D2100000 00000000
00002234  4483F287 00000000                       1639          DC    XL16'4483F28700000000 0000000000000000'
0000223C  00000000 00000000
00002244  C483F287 00000000                       1640          DC    XL16'C483F28700000000 0000000000000000'
0000224C  00000000 00000000
                                                  1641
                                                  1642 * +Zero, -Zero
                                                  1643          VRR_C VCRNF,0,2,00,00,N
00002258                                          1644+         DS    0FD
00002258                        00002258          1645+         USING *,R5              base for test data and test routine
00002258  000022F8                                1646+T18      DC    A(X18)            address of test routine
0000225C  0012                                    1647+         DC    H'18'             test number
0000225E  00                                      1648+         DC    X'00'
0000225F  D5                                      1649+         DC    CL1'N'            Y = skip cross check
00002260  00                                      1650+         DC    HL1'0'            m4
00002261  02                                      1651+         DC    HL1'2'            m5
00002262  00                                      1652+FLG18    DC    X'00'             expected FPC flags
00002263  00                                      1653+VXC18    DC    X'00'             expected VXC
00002264  0000232C                                1654+V2_18    DC    A(RE18+16)        address of v2: 16-byte packed decimal
00002268  E5C3D9D5 C6404040                       1655+         DC    CL8'VCRNF'        instruction name
00002270  00000010                                1656+         DC    A(16)             result length
00002274  0000231C                                1657+         DC    A(RE18)           address of expected resul
00002278  00000000 00000000                       1658+         DS    FD                gap
00002280  00000000 00000000                       1659+V1O18    DS    XL16              V1 output
00002288  00000000 00000000
00002290  00000000 00000000                       1660+         DS    FD                   gap
00002298  00000000                                1661+FPC_R_18 DS    F                 FPC after instruction
000022A0  00000000 00000000                       1662+         DS    FD                   gap
000022A8  40404040 40404040                       1663+         DC    CL8' '            was cross check skipped?
000022B0  00000000 00000000                       1664+         DS    FD                   gap
000022B8  00000000 00000000                       1665+XCO18    DS    XL16              Cross check Output
000022C0  00000000 00000000
000022C8  00000000 00000000                       1666+         DS    XL16
000022D0  00000000 00000000
000022D8  00000000 00000000                       1667+         DS    FD                   gap
000022E0  00000000                                1668+FPC_XC_18 DS   F                1st cross check FPC
000022E4  00000000                                1669+         DS    F                2nd cross check FPC
000022E8  00000000 00000000                       1670+         DS    FD                   gap
000022F0  00000000                                1671+         DS    F                 debug area
000022F4  00000000                                1672+         DS    F
                                                  1673+*
000022F8                                          1674+X18      DS    0F
000022F8  B29D 852C                     0000072C  1675+         LFPC  FPCMOFF           initialize FPC
000022FC  E320 500C 0014                00002264  1676+         LGF   R2,V2_18          get v2
00002302  E767 2000 0C36                00000000  1677+         VLM   V22,V23,0(R2)
00002308  E666 7002 0E75                          1678+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
0000230E  B29C 5040                     00002298  1679+         STFPC FPC_R_18          save FPC
00002312  E760 5028 080E                00002280  1680+         VST   V22,V1O18         save instruction result
00002318  07FB                                    1681+         BR    R11               return
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0000231C                                           1682+RE18     DS    0F                expected 16 byte result
0000231C                                           1683+         DROP  R5
0000231C  00000000 00000000                        1684          DC    XL16'0000000000000000 8000000000000000'
00002324  80000000 00000000
0000232C  00000000 00000000                        1685          DC    XL16'0000000000000000 0000000000000000'
00002334  00000000 00000000
0000233C  80000000 00000000                        1686          DC    XL16'8000000000000000 0000000000000000'
00002344  00000000 00000000
                                                   1687
                                                   1688 * +Normal, -Normal
                                                   1689          VRR_C VCRNF,0,2,00,00,N                skip xc: lost digits
00002350                                           1690+         DS    0FD
00002350                      00002350             1691+         USING *,R5              base for test data and test routine
00002350  000023F0                                 1692+T19      DC    A(X19)            address of test routine
00002354  0013                                     1693+         DC    H'19'             test number
00002356  00                                       1694+         DC    X'00'
00002357  D5                                       1695+         DC    CL1'N'            Y = skip cross check
00002358  00                                       1696+         DC    HL1'0'            m4
00002359  02                                       1697+         DC    HL1'2'            m5
0000235A  00                                       1698+FLG19    DC    X'00'             expected FPC flags
0000235B  00                                       1699+VXC19    DC    X'00'             expected VXC
0000235C  00002424                                 1700+V2_19    DC    A(RE19+16)        address of v2: 16-byte packed decimal
00002360  E5C3D9D5 C6404040                        1701+         DC    CL8'VCRNF'        instruction name
00002368  00000010                                 1702+         DC    A(16)             result length
0000236C  00002414                                 1703+         DC    A(RE19)           address of expected resul
00002370  00000000 00000000                        1704+         DS    FD                gap
00002378  00000000 00000000                        1705+V1O19    DS    XL16              V1 output
00002380  00000000 00000000
00002388  00000000 00000000                        1706+         DS    FD                   gap
00002390  00000000                                 1707+FPC_R_19 DS    F                 FPC after instruction
00002398  00000000 00000000                        1708+         DS    FD                   gap
000023A0  40404040 40404040                        1709+         DC    CL8' '            was cross check skipped?
000023A8  00000000 00000000                        1710+         DS    FD                   gap
000023B0  00000000 00000000                        1711+XCO19    DS    XL16              Cross check Output
000023B8  00000000 00000000
000023C0  00000000 00000000                        1712+         DS    XL16
000023C8  00000000 00000000
000023D0  00000000 00000000                        1713+         DS    FD                   gap
000023D8  00000000                                 1714+FPC_XC_19 DS   F                1st cross check FPC
000023DC  00000000                                 1715+         DS    F                2nd cross check FPC
000023E0  00000000 00000000                        1716+         DS    FD                   gap
000023E8  00000000                                 1717+         DS    F                 debug area
000023EC  00000000                                 1718+         DS    F
                                                   1719+*
000023F0                                           1720+X19      DS    0F
000023F0  B29D 852C                   0000072C     1721+         LFPC  FPCMOFF           initialize FPC
000023F4  E320 500C 0014              0000235C     1722+         LGF   R2,V2_19          get v2
000023FA  E767 2000 0C36              00000000     1723+         VLM   V22,V23,0(R2)
00002400  E666 7002 0E75                           1724+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
00002406  B29C 5040                   00002390     1725+         STFPC FPC_R_19          save FPC
0000240A  E760 5028 080E              00002378     1726+         VST   V22,V1O19         save instruction result
00002410  07FB                                     1727+         BR    R11               return
00002414                                           1728+RE19     DS    0F                expected 16 byte result
00002414                                           1729+         DROP  R5
00002414  40800000 00000000                        1730          DC    XL16'4080000000000000 C080000000000000'
0000241C  C0800000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002424  40200000 00000000                        1731          DC    XL16'4020000000000000 0000000000000000'
0000242C  00000000 00000000
00002434  C0200000 00000000                        1732          DC    XL16'C020000000000000 0000000000000000'
0000243C  00000000 00000000
                                                   1733
                                                   1734 * +Subnormal, -Subnormal
                                                   1735          VRR_C VCRNF,0,2,10,00,S                skip xc: lost digits
00002448                                           1736+         DS    0FD
00002448                       00002448            1737+         USING *,R5              base for test data and test routine
00002448  000024E8                                 1738+T20      DC    A(X20)            address of test routine
0000244C  0014                                     1739+         DC    H'20'             test number
0000244E  00                                       1740+         DC    X'00'
0000244F  E2                                       1741+         DC    CL1'S'            Y = skip cross check
00002450  00                                       1742+         DC    HL1'0'            m4
00002451  02                                       1743+         DC    HL1'2'            m5
00002452  10                                       1744+FLG20    DC    X'10'             expected FPC flags
00002453  00                                       1745+VXC20    DC    X'00'             expected VXC
00002454  0000251C                                 1746+V2_20    DC    A(RE20+16)        address of v2: 16-byte packed decimal
00002458  E5C3D9D5 C6404040                        1747+         DC    CL8'VCRNF'        instruction name
00002460  00000010                                 1748+         DC    A(16)             result length
00002464  0000250C                                 1749+         DC    A(RE20)           address of expected resul
00002468  00000000 00000000                        1750+         DS    FD                gap
00002470  00000000 00000000                        1751+V1O20    DS    XL16              V1 output
00002478  00000000 00000000
00002480  00000000 00000000                        1752+         DS    FD                   gap
00002488  00000000                                 1753+FPC_R_20 DS    F                 FPC after instruction
00002490  00000000 00000000                        1754+         DS    FD                   gap
00002498  40404040 40404040                        1755+         DC    CL8' '            was cross check skipped?
000024A0  00000000 00000000                        1756+         DS    FD                   gap
000024A8  00000000 00000000                        1757+XCO20    DS    XL16              Cross check Output
000024B0  00000000 00000000
000024B8  00000000 00000000                        1758+         DS    XL16
000024C0  00000000 00000000
000024C8  00000000 00000000                        1759+         DS    FD                   gap
000024D0  00000000                                 1760+FPC_XC_20 DS   F                1st cross check FPC
000024D4  00000000                                 1761+         DS    F                2nd cross check FPC
000024D8  00000000 00000000                        1762+         DS    FD                   gap
000024E0  00000000                                 1763+         DS    F                 debug area
000024E4  00000000                                 1764+         DS    F
                                                   1765+*
000024E8                                           1766+X20      DS    0F
000024E8  B29D 852C                    0000072C    1767+         LFPC  FPCMOFF           initialize FPC
000024EC  E320 500C 0014               00002454    1768+         LGF   R2,V2_20          get v2
000024F2  E767 2000 0C36               00000000    1769+         VLM   V22,V23,0(R2)
000024F8  E666 7002 0E75                           1770+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
000024FE  B29C 5040                    00002488    1771+         STFPC FPC_R_20          save FPC
00002502  E760 5028 080E               00002470    1772+         VST   V22,V1O20         save instruction result
00002508  07FB                                     1773+         BR    R11               return
0000250C                                           1774+RE20     DS    0F                expected 16 byte result
0000250C                                           1775+         DROP  R5
0000250C  00000000 00000000                        1776          DC    XL16'0000000000000000 8000000000000000'
00002514  80000000 00000000
0000251C  007FFFFF 00000000                        1777          DC    XL16'007FFFFF00000000 0000000000000000'
00002524  00000000 00000000
0000252C  807FFFFF 00000000                        1778          DC    XL16'807FFFFF00000000 0000000000000000'
00002534  00000000 00000000
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

                                                   1779
                                                   1780 * +Infinity, -Infinity
                                                   1781          VRR_C VCRNF,0,2,20,00,S                skip xc: lost digits
00002540                                           1782+         DS    0FD
00002540                       00002540            1783+         USING *,R5              base for test data and test routine
00002540  000025E0                                 1784+T21      DC    A(X21)            address of test routine
00002544  0015                                     1785+         DC    H'21'             test number
00002546  00                                       1786+         DC    X'00'
00002547  E2                                       1787+         DC    CL1'S'            Y = skip cross check
00002548  00                                       1788+         DC    HL1'0'            m4
00002549  02                                       1789+         DC    HL1'2'            m5
0000254A  20                                       1790+FLG21    DC    X'20'             expected FPC flags
0000254B  00                                       1791+VXC21    DC    X'00'             expected VXC
0000254C  00002614                                 1792+V2_21    DC    A(RE21+16)        address of v2: 16-byte packed decimal
00002550  E5C3D9D5 C6404040                        1793+         DC    CL8'VCRNF'        instruction name
00002558  00000010                                 1794+         DC    A(16)             result length
0000255C  00002604                                 1795+         DC    A(RE21)           address of expected resul
00002560  00000000 00000000                        1796+         DS    FD                gap
00002568  00000000 00000000                        1797+V1O21    DS    XL16              V1 output
00002570  00000000 00000000
00002578  00000000 00000000                        1798+         DS    FD                   gap
00002580  00000000                                 1799+FPC_R_21 DS    F                 FPC after instruction
00002588  00000000 00000000                        1800+         DS    FD                   gap
00002590  40404040 40404040                        1801+         DC    CL8' '            was cross check skipped?
00002598  00000000 00000000                        1802+         DS    FD                   gap
000025A0  00000000 00000000                        1803+XCO21    DS    XL16              Cross check Output
000025A8  00000000 00000000
000025B0  00000000 00000000                        1804+         DS    XL16
000025B8  00000000 00000000
000025C0  00000000 00000000                        1805+         DS    FD                   gap
000025C8  00000000                                 1806+FPC_XC_21 DS   F                1st cross check FPC
000025CC  00000000                                 1807+         DS    F                2nd cross check FPC
000025D0  00000000 00000000                        1808+         DS    FD                   gap
000025D8  00000000                                 1809+         DS    F                 debug area
000025DC  00000000                                 1810+         DS    F
                                                   1811+*
000025E0                                           1812+X21      DS    0F
000025E0  B29D 852C                     0000072C   1813+         LFPC  FPCMOFF           initialize FPC
000025E4  E320 500C 0014                0000254C   1814+         LGF   R2,V2_21          get v2
000025EA  E767 2000 0C36                00000000   1815+         VLM   V22,V23,0(R2)
000025F0  E666 7002 0E75                           1816+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
000025F6  B29C 5040                     00002580   1817+         STFPC FPC_R_21          save FPC
000025FA  E760 5028 080E                00002568   1818+         VST   V22,V1O21         save instruction result
00002600  07FB                                     1819+         BR    R11               return
00002604                                           1820+RE21     DS    0F                expected 16 byte result
00002604                                           1821+         DROP  R5
00002604  7FFF0000 00000000                        1822          DC    XL16'7FFF000000000000 FFFF000000000000'
0000260C  FFFF0000 00000000
00002614  7F800000 00000000                        1823          DC    XL16'7F80000000000000 0000000000000000'
0000261C  00000000 00000000
00002624  FF800000 00000000                        1824          DC    XL16'FF80000000000000 0000000000000000'
0000262C  00000000 00000000
                                                   1825
                                                   1826 * +QNaN, -QNaN
                                                   1827          VRR_C VCRNF,0,2,80,00,S                skip xc: lost digits
00002638                                           1828+         DS    0FD
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002638                     00002638             1829+         USING *,R5              base for test data and test routine
00002638  000026D8                                1830+T22      DC    A(X22)            address of test routine
0000263C  0016                                    1831+         DC    H'22'             test number
0000263E  00                                      1832+         DC    X'00'
0000263F  E2                                      1833+         DC    CL1'S'            Y = skip cross check
00002640  00                                      1834+         DC    HL1'0'            m4
00002641  02                                      1835+         DC    HL1'2'            m5
00002642  80                                      1836+FLG22    DC    X'80'             expected FPC flags
00002643  00                                      1837+VXC22    DC    X'00'             expected VXC
00002644  0000270C                                1838+V2_22    DC    A(RE22+16)        address of v2: 16-byte packed decimal
00002648  E5C3D9D5 C6404040                       1839+         DC    CL8'VCRNF'        instruction name
00002650  00000010                                1840+         DC    A(16)             result length
00002654  000026FC                                1841+         DC    A(RE22)           address of expected resul
00002658  00000000 00000000                       1842+         DS    FD                gap
00002660  00000000 00000000                       1843+V1O22    DS    XL16              V1 output
00002668  00000000 00000000
00002670  00000000 00000000                       1844+         DS    FD                   gap
00002678  00000000                                1845+FPC_R_22 DS    F                 FPC after instruction
00002680  00000000 00000000                       1846+         DS    FD                   gap
00002688  40404040 40404040                       1847+         DC    CL8' '            was cross check skipped?
00002690  00000000 00000000                       1848+         DS    FD                   gap
00002698  00000000 00000000                       1849+XCO22    DS    XL16              Cross check Output
000026A0  00000000 00000000
000026A8  00000000 00000000                       1850+         DS    XL16
000026B0  00000000 00000000
000026B8  00000000 00000000                       1851+         DS    FD                   gap
000026C0  00000000                                1852+FPC_XC_22 DS   F                1st cross check FPC
000026C4  00000000                                1853+         DS    F                2nd cross check FPC
000026C8  00000000 00000000                       1854+         DS    FD                   gap
000026D0  00000000                                1855+         DS    F                 debug area
000026D4  00000000                                1856+         DS    F
                                                  1857+*
000026D8                                          1858+X22      DS    0F
000026D8  B29D 852C                    0000072C   1859+         LFPC  FPCMOFF           initialize FPC
000026DC  E320 500C 0014               00002644   1860+         LGF   R2,V2_22          get v2
000026E2  E767 2000 0C36               00000000   1861+         VLM   V22,V23,0(R2)
000026E8  E666 7002 0E75                          1862+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
000026EE  B29C 5040                    00002678   1863+         STFPC FPC_R_22          save FPC
000026F2  E760 5028 080E               00002660   1864+         VST   V22,V1O22         save instruction result
000026F8  07FB                                    1865+         BR    R11               return
000026FC                                          1866+RE22     DS    0F                expected 16 byte result
000026FC                                          1867+         DROP  R5
000026FC  7FFF0000 00000000                       1868          DC    XL16'7FFF000000000000 FFFF000000000000'
00002704  FFFF0000 00000000
0000270C  7FC10000 00000000                       1869          DC    XL16'7FC1000000000000 0000000000000000'
00002714  00000000 00000000
0000271C  FFC10000 00000000                       1870          DC    XL16'FFC1000000000000 0000000000000000'
00002724  00000000 00000000
                                                  1871
                                                  1872 * +SNaN, -SNaN
                                                  1873          VRR_C VCRNF,0,2,80,00,S                 skip xc: lost digits
00002730                                          1874+         DS    0FD
00002730                     00002730             1875+         USING *,R5              base for test data and test routine
00002730  000027D0                                1876+T23      DC    A(X23)            address of test routine
00002734  0017                                    1877+         DC    H'23'             test number
00002736  00                                      1878+         DC    X'00'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002737  E2                                       1879+         DC    CL1'S'            Y = skip cross check
00002738  00                                       1880+         DC    HL1'0'            m4
00002739  02                                       1881+         DC    HL1'2'            m5
0000273A  80                                       1882+FLG23    DC    X'80'             expected FPC flags
0000273B  00                                       1883+VXC23    DC    X'00'             expected VXC
0000273C  00002804                                 1884+V2_23    DC    A(RE23+16)        address of v2: 16-byte packed decimal
00002740  E5C3D9D5 C6404040                        1885+         DC    CL8'VCRNF'        instruction name
00002748  00000010                                 1886+         DC    A(16)             result length
0000274C  000027F4                                 1887+         DC    A(RE23)           address of expected resul
00002750  00000000 00000000                        1888+         DS    FD                gap
00002758  00000000 00000000                        1889+V1O23    DS    XL16              V1 output
00002760  00000000 00000000
00002768  00000000 00000000                        1890+         DS    FD                   gap
00002770  00000000                                 1891+FPC_R_23 DS    F                 FPC after instruction
00002778  00000000 00000000                        1892+         DS    FD                   gap
00002780  40404040 40404040                        1893+         DC    CL8' '            was cross check skipped?
00002788  00000000 00000000                        1894+         DS    FD                   gap
00002790  00000000 00000000                        1895+XCO23    DS    XL16              Cross check Output
00002798  00000000 00000000
000027A0  00000000 00000000                        1896+         DS    XL16
000027A8  00000000 00000000
000027B0  00000000 00000000                        1897+         DS    FD                   gap
000027B8  00000000                                 1898+FPC_XC_23 DS   F                1st cross check FPC
000027BC  00000000                                 1899+         DS    F                2nd cross check FPC
000027C0  00000000 00000000                        1900+         DS    FD                   gap
000027C8  00000000                                 1901+         DS    F                 debug area
000027CC  00000000                                 1902+         DS    F
                                                   1903+*
000027D0                                           1904+X23      DS    0F
000027D0  B29D 852C                     0000072C   1905+         LFPC  FPCMOFF           initialize FPC
000027D4  E320 500C 0014                0000273C   1906+         LGF   R2,V2_23          get v2
000027DA  E767 2000 0C36                00000000   1907+         VLM   V22,V23,0(R2)
000027E0  E666 7002 0E75                           1908+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
000027E6  B29C 5040                     00002770   1909+         STFPC FPC_R_23          save FPC
000027EA  E760 5028 080E                00002758   1910+         VST   V22,V1O23         save instruction result
000027F0  07FB                                     1911+         BR    R11               return
000027F4                                           1912+RE23     DS    0F                expected 16 byte result
000027F4                                           1913+         DROP  R5
000027F4  7FFF0000 00000000                        1914          DC    XL16'7FFF000000000000 FFFF000000000000'
000027FC  FFFF0000 00000000
00002804  7F810000 00000000                        1915          DC    XL16'7F81000000000000 0000000000000000'
0000280C  00000000 00000000
00002814  FF810000 00000000                        1916          DC    XL16'FF81000000000000 0000000000000000'
0000281C  00000000 00000000
                                                   1917
                                                   1918 * Additional tests - multiple excecptions
                                                   1919 *-----------------------------------------------------------------
                                                   1920
                                                   1921 * +Subnormal, -Infinity
                                                   1922          VRR_C VCRNF,0,2,30,00,S                 skip xc: lost digits
00002828                                           1923+         DS    0FD
00002828                        00002828           1924+         USING *,R5              base for test data and test routine
00002828  000028C8                                 1925+T24      DC    A(X24)            address of test routine
0000282C  0018                                     1926+         DC    H'24'             test number
0000282E  00                                       1927+         DC    X'00'
0000282F  E2                                       1928+         DC    CL1'S'            Y = skip cross check
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002830  00                                       1929+         DC    HL1'0'            m4
00002831  02                                       1930+         DC    HL1'2'            m5
00002832  30                                       1931+FLG24    DC    X'30'             expected FPC flags
00002833  00                                       1932+VXC24    DC    X'00'             expected VXC
00002834  000028FC                                 1933+V2_24    DC    A(RE24+16)        address of v2: 16-byte packed decimal
00002838  E5C3D9D5 C6404040                        1934+         DC    CL8'VCRNF'        instruction name
00002840  00000010                                 1935+         DC    A(16)             result length
00002844  000028EC                                 1936+         DC    A(RE24)           address of expected resul
00002848  00000000 00000000                        1937+         DS    FD                gap
00002850  00000000 00000000                        1938+V1O24    DS    XL16              V1 output
00002858  00000000 00000000
00002860  00000000 00000000                        1939+         DS    FD                   gap
00002868  00000000                                 1940+FPC_R_24 DS    F                 FPC after instruction
00002870  00000000 00000000                        1941+         DS    FD                   gap
00002878  40404040 40404040                        1942+         DC    CL8' '            was cross check skipped?
00002880  00000000 00000000                        1943+         DS    FD                   gap
00002888  00000000 00000000                        1944+XCO24    DS    XL16              Cross check Output
00002890  00000000 00000000
00002898  00000000 00000000                        1945+         DS    XL16
000028A0  00000000 00000000
000028A8  00000000 00000000                        1946+         DS    FD                   gap
000028B0  00000000                                 1947+FPC_XC_24 DS   F                1st cross check FPC
000028B4  00000000                                 1948+         DS    F                2nd cross check FPC
000028B8  00000000 00000000                        1949+         DS    FD                   gap
000028C0  00000000                                 1950+         DS    F                 debug area
000028C4  00000000                                 1951+         DS    F
                                                   1952+*
000028C8                                           1953+X24      DS    0F
000028C8  B29D 852C                     0000072C   1954+         LFPC  FPCMOFF           initialize FPC
000028CC  E320 500C 0014                00002834   1955+         LGF   R2,V2_24          get v2
000028D2  E767 2000 0C36                00000000   1956+         VLM   V22,V23,0(R2)
000028D8  E666 7002 0E75                           1957+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
000028DE  B29C 5040                     00002868   1958+         STFPC FPC_R_24          save FPC
000028E2  E760 5028 080E                00002850   1959+         VST   V22,V1O24         save instruction result
000028E8  07FB                                     1960+         BR    R11               return
000028EC                                           1961+RE24     DS    0F                expected 16 byte result
000028EC                                           1962+         DROP  R5
000028EC  00000000 00000000                        1963          DC    XL16'0000000000000000 FFFF000000000000'
000028F4  FFFF0000 00000000
000028FC  007FFFFF 00000000                        1964          DC    XL16'007FFFFF00000000 0000000000000000'
00002904  00000000 00000000
0000290C  FF800000 00000000                        1965          DC    XL16'FF80000000000000 0000000000000000'
00002914  00000000 00000000
                                                   1966
                                                   1967 * -Infinity, +Subnormal
                                                   1968          VRR_C VCRNF,0,2,30,00,S                skip xc: lost digits
00002920                                           1969+         DS    0FD
00002920                        00002920           1970+         USING *,R5              base for test data and test routine
00002920  000029C0                                 1971+T25      DC    A(X25)            address of test routine
00002924  0019                                     1972+         DC    H'25'             test number
00002926  00                                       1973+         DC    X'00'
00002927  E2                                       1974+         DC    CL1'S'            Y = skip cross check
00002928  00                                       1975+         DC    HL1'0'            m4
00002929  02                                       1976+         DC    HL1'2'            m5
0000292A  30                                       1977+FLG25    DC    X'30'             expected FPC flags
0000292B  00                                       1978+VXC25    DC    X'00'             expected VXC
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

0000292C  000029F4                                 1979+V2_25    DC    A(RE25+16)        address of v2: 16-byte packed decimal
00002930  E5C3D9D5 C6404040                        1980+         DC    CL8'VCRNF'        instruction name
00002938  00000010                                 1981+         DC    A(16)             result length
0000293C  000029E4                                 1982+         DC    A(RE25)           address of expected resul
00002940  00000000 00000000                        1983+         DS    FD                gap
00002948  00000000 00000000                        1984+V1O25    DS    XL16              V1 output
00002950  00000000 00000000
00002958  00000000 00000000                        1985+         DS    FD                   gap
00002960  00000000                                 1986+FPC_R_25 DS    F                 FPC after instruction
00002968  00000000 00000000                        1987+         DS    FD                   gap
00002970  40404040 40404040                        1988+         DC    CL8' '            was cross check skipped?
00002978  00000000 00000000                        1989+         DS    FD                   gap
00002980  00000000 00000000                        1990+XCO25    DS    XL16              Cross check Output
00002988  00000000 00000000
00002990  00000000 00000000                        1991+         DS    XL16
00002998  00000000 00000000
000029A0  00000000 00000000                        1992+         DS    FD                   gap
000029A8  00000000                                 1993+FPC_XC_25 DS   F                1st cross check FPC
000029AC  00000000                                 1994+         DS    F                2nd cross check FPC
000029B0  00000000 00000000                        1995+         DS    FD                   gap
000029B8  00000000                                 1996+         DS    F                 debug area
000029BC  00000000                                 1997+         DS    F
                                                   1998+*
000029C0                                           1999+X25      DS    0F
000029C0  B29D 852C                     0000072C   2000+         LFPC  FPCMOFF           initialize FPC
000029C4  E320 500C 0014                0000292C   2001+         LGF   R2,V2_25          get v2
000029CA  E767 2000 0C36                00000000   2002+         VLM   V22,V23,0(R2)
000029D0  E666 7002 0E75                           2003+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
000029D6  B29C 5040                     00002960   2004+         STFPC FPC_R_25          save FPC
000029DA  E760 5028 080E                00002948   2005+         VST   V22,V1O25         save instruction result
000029E0  07FB                                     2006+         BR    R11               return
000029E4                                           2007+RE25     DS    0F                expected 16 byte result
000029E4                                           2008+         DROP  R5
000029E4  00000000 00000000                        2009          DC    XL16'0000000000000000 FFFF000000000000'
000029EC  FFFF0000 00000000
000029F4  00000000 00000000                        2010          DC    XL16'0000000000000000 0000000000000000'
000029FC  00000000 00000000
00002A04  FF800000 00000000                        2011          DC    XL16'FF80000000000000 007FFFFF00000000'
00002A0C  007FFFFF 00000000
                                                   2012
                                                   2013 * +Infinity, -QNaN
                                                   2014          VRR_C VCRNF,0,2,A0,00,S                 skip xc: lost digits
00002A18                                           2015+         DS    0FD
00002A18                        00002A18           2016+         USING *,R5              base for test data and test routine
00002A18  00002AB8                                 2017+T26      DC    A(X26)            address of test routine
00002A1C  001A                                     2018+         DC    H'26'             test number
00002A1E  00                                       2019+         DC    X'00'
00002A1F  E2                                       2020+         DC    CL1'S'            Y = skip cross check
00002A20  00                                       2021+         DC    HL1'0'            m4
00002A21  02                                       2022+         DC    HL1'2'            m5
00002A22  A0                                       2023+FLG26    DC    X'A0'             expected FPC flags
00002A23  00                                       2024+VXC26    DC    X'00'             expected VXC
00002A24  00002AEC                                 2025+V2_26    DC    A(RE26+16)        address of v2: 16-byte packed decimal
00002A28  E5C3D9D5 C6404040                        2026+         DC    CL8'VCRNF'        instruction name
00002A30  00000010                                 2027+         DC    A(16)             result length
00002A34  00002ADC                                 2028+         DC    A(RE26)           address of expected resul
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002A38  00000000 00000000                        2029+          DS    FD                 gap
00002A40  00000000 00000000                        2030+V1O26     DS    XL16               V1 output
00002A48  00000000 00000000
00002A50  00000000 00000000                        2031+          DS    FD                    gap
00002A58  00000000                                 2032+FPC_R_26  DS    F                  FPC after instruction
00002A60  00000000 00000000                        2033+          DS    FD                    gap
00002A68  40404040 40404040                        2034+          DC    CL8' '             was cross check skipped?
00002A70  00000000 00000000                        2035+          DS    FD                    gap
00002A78  00000000 00000000                        2036+XCO26     DS    XL16               Cross check Output
00002A80  00000000 00000000
00002A88  00000000 00000000                        2037+          DS    XL16
00002A90  00000000 00000000
00002A98  00000000 00000000                        2038+          DS    FD                    gap
00002AA0  00000000                                 2039+FPC_XC_26 DS    F                 1st cross check FPC
00002AA4  00000000                                 2040+          DS    F                 2nd cross check FPC
00002AA8  00000000 00000000                        2041+          DS    FD                    gap
00002AB0  00000000                                 2042+          DS    F                  debug area
00002AB4  00000000                                 2043+          DS    F
                                                   2044+*
00002AB8                                           2045+X26       DS    0F
00002AB8  B29D 852C                     0000072C   2046+          LFPC  FPCMOFF            initialize FPC
00002ABC  E320 500C 0014                00002A24   2047+          LGF   R2,V2_26           get v2
00002AC2  E767 2000 0C36                00000000   2048+          VLM   V22,V23,0(R2)
00002AC8  E666 7002 0E75                           2049+          VCRNF V22,V22,V23,0,2       test instruction (dest is source)
00002ACE  B29C 5040                     00002A58   2050+          STFPC FPC_R_26           save FPC
00002AD2  E760 5028 080E                00002A40   2051+          VST   V22,V1O26          save instruction result
00002AD8  07FB                                     2052+          BR    R11                return
00002ADC                                           2053+RE26      DS    0F                 expected 16 byte result
00002ADC                                           2054+          DROP  R5
00002ADC  7FFF0000 00000000                        2055           DC    XL16'7FFF000000000000 FFFF000000000000'
00002AE4  FFFF0000 00000000
00002AEC  7F800000 00000000                        2056           DC    XL16'7F80000000000000 0000000000000000'
00002AF4  00000000 00000000
00002AFC  FFC10000 00000000                        2057           DC    XL16'FFC1000000000000 0000000000000000'
00002B04  00000000 00000000
                                                   2058
                                                   2059 * +Infinity, -QNaN, +Infinity
                                                   2060           VRR_C VCRNF,0,2,A0,00,S                 skip xc: lost digits
00002B10                                           2061+          DS    0FD
00002B10                        00002B10           2062+          USING *,R5               base for test data and test routine
00002B10  00002BB0                                 2063+T27       DC    A(X27)             address of test routine
00002B14  001B                                     2064+          DC    H'27'              test number
00002B16  00                                       2065+          DC    X'00'
00002B17  E2                                       2066+          DC    CL1'S'             Y = skip cross check
00002B18  00                                       2067+          DC    HL1'0'             m4
00002B19  02                                       2068+          DC    HL1'2'             m5
00002B1A  A0                                       2069+FLG27     DC    X'A0'              expected FPC flags
00002B1B  00                                       2070+VXC27     DC    X'00'              expected VXC
00002B1C  00002BE4                                 2071+V2_27     DC    A(RE27+16)         address of v2: 16-byte packed decimal
00002B20  E5C3D9D5 C6404040                        2072+          DC    CL8'VCRNF'         instruction name
00002B28  00000010                                 2073+          DC    A(16)              result length
00002B2C  00002BD4                                 2074+          DC    A(RE27)            address of expected resul
00002B30  00000000 00000000                        2075+          DS    FD                 gap
00002B38  00000000 00000000                        2076+V1O27     DS    XL16               V1 output
00002B40  00000000 00000000
00002B48  00000000 00000000                        2077+          DS    FD                    gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002B50  00000000                                 2078+FPC_R_27 DS    F                 FPC after instruction
00002B58  00000000 00000000                        2079+         DS    FD                   gap
00002B60  40404040 40404040                        2080+         DC    CL8' '            was cross check skipped?
00002B68  00000000 00000000                        2081+         DS    FD                   gap
00002B70  00000000 00000000                        2082+XCO27    DS    XL16              Cross check Output
00002B78  00000000 00000000
00002B80  00000000 00000000                        2083+         DS    XL16
00002B88  00000000 00000000
00002B90  00000000 00000000                        2084+         DS    FD                   gap
00002B98  00000000                                 2085+FPC_XC_27 DS   F                1st cross check FPC
00002B9C  00000000                                 2086+         DS    F                2nd cross check FPC
00002BA0  00000000 00000000                        2087+         DS    FD                   gap
00002BA8  00000000                                 2088+         DS    F                 debug area
00002BAC  00000000                                 2089+         DS    F
                                                   2090+*
00002BB0                                           2091+X27      DS    0F
00002BB0  B29D 852C                     0000072C   2092+         LFPC  FPCMOFF           initialize FPC
00002BB4  E320 500C 0014                00002B1C   2093+         LGF   R2,V2_27          get v2
00002BBA  E767 2000 0C36                00000000   2094+         VLM   V22,V23,0(R2)
00002BC0  E666 7002 0E75                           2095+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
00002BC6  B29C 5040                     00002B50   2096+         STFPC FPC_R_27          save FPC
00002BCA  E760 5028 080E                00002B38   2097+         VST   V22,V1O27         save instruction result
00002BD0  07FB                                     2098+         BR    R11               return
00002BD4                                           2099+RE27     DS    0F                expected 16 byte result
00002BD4                                           2100+         DROP  R5
00002BD4  7FFF0000 00000000                        2101          DC    XL16'7FFF000000000000 FFFF00007FFF0000'
00002BDC  FFFF0000 7FFF0000
00002BE4  7F800000 00000000                        2102          DC    XL16'7F80000000000000 0000000000000000'
00002BEC  00000000 00000000
00002BF4  FFC10000 00000000                        2103          DC    XL16'FFC1000000000000 7F80000000000000'
00002BFC  7F800000 00000000
                                                   2104
                                                   2105 * +Infinity, -QNaN, +Subnormal
                                                   2106          VRR_C VCRNF,0,2,B0,00,S                skip xc: lost digits
00002C08                                           2107+         DS    0FD
00002C08                        00002C08           2108+         USING *,R5              base for test data and test routine
00002C08  00002CA8                                 2109+T28      DC    A(X28)            address of test routine
00002C0C  001C                                     2110+         DC    H'28'             test number
00002C0E  00                                       2111+         DC    X'00'
00002C0F  E2                                       2112+         DC    CL1'S'            Y = skip cross check
00002C10  00                                       2113+         DC    HL1'0'            m4
00002C11  02                                       2114+         DC    HL1'2'            m5
00002C12  B0                                       2115+FLG28    DC    X'B0'             expected FPC flags
00002C13  00                                       2116+VXC28    DC    X'00'             expected VXC
00002C14  00002CDC                                 2117+V2_28    DC    A(RE28+16)        address of v2: 16-byte packed decimal
00002C18  E5C3D9D5 C6404040                        2118+         DC    CL8'VCRNF'        instruction name
00002C20  00000010                                 2119+         DC    A(16)             result length
00002C24  00002CCC                                 2120+         DC    A(RE28)           address of expected resul
00002C28  00000000 00000000                        2121+         DS    FD                gap
00002C30  00000000 00000000                        2122+V1O28    DS    XL16              V1 output
00002C38  00000000 00000000
00002C40  00000000 00000000                        2123+         DS    FD                   gap
00002C48  00000000                                 2124+FPC_R_28 DS    F                 FPC after instruction
00002C50  00000000 00000000                        2125+         DS    FD                   gap
00002C58  40404040 40404040                        2126+         DC    CL8' '            was cross check skipped?
00002C60  00000000 00000000                        2127+         DS    FD                   gap
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002C68  00000000 00000000                        2128+XCO28    DS    XL16              Cross check Output
00002C70  00000000 00000000
00002C78  00000000 00000000                        2129+         DS    XL16
00002C80  00000000 00000000
00002C88  00000000 00000000                        2130+         DS    FD                   gap
00002C90  00000000                                 2131+FPC_XC_28 DS   F                1st cross check FPC
00002C94  00000000                                 2132+         DS    F                2nd cross check FPC
00002C98  00000000 00000000                        2133+         DS    FD                   gap
00002CA0  00000000                                 2134+         DS    F                 debug area
00002CA4  00000000                                 2135+         DS    F
                                                   2136+*
00002CA8                                           2137+X28      DS    0F
00002CA8  B29D 852C                    0000072C    2138+         LFPC  FPCMOFF           initialize FPC
00002CAC  E320 500C 0014               00002C14    2139+         LGF   R2,V2_28          get v2
00002CB2  E767 2000 0C36               00000000    2140+         VLM   V22,V23,0(R2)
00002CB8  E666 7002 0E75                           2141+         VCRNF V22,V22,V23,0,2       test instruction (dest is source)
00002CBE  B29C 5040                    00002C48    2142+         STFPC FPC_R_28          save FPC
00002CC2  E760 5028 080E               00002C30    2143+         VST   V22,V1O28         save instruction result
00002CC8  07FB                                     2144+         BR    R11               return
00002CCC                                           2145+RE28     DS    0F                expected 16 byte result
00002CCC                                           2146+         DROP  R5
00002CCC  7FFF0000 00000000                        2147          DC    XL16'7FFF000000000000 FFFF000000000000'
00002CD4  FFFF0000 00000000
00002CDC  7F800000 00000000                        2148          DC    XL16'7F8000000000000 0000000000000000'
00002CE4  00000000 00000000
00002CEC  FFC10000 00000000                        2149          DC    XL16'FFC1000000000000 007FFFFF00000000'
00002CF4  007FFFFF 00000000
                                                   2150
                                                   2151
00002CFC  00000000                                 2152          DC    F'0'     END OF TABLE
00002D00  00000000                                 2153          DC    F'0'
                                                   2154 *
                                                   2155 * table of pointers to individual tests
                                                   2156 *
00002D04                                           2157 E6TESTS  DS    0F
                                                   2158          PTTABLE
00002D04                                           2159+TTABLE   DS    0F
00002D04  000011E0                                 2160+         DC    A(T1)             TEST &CUR
00002D08  000012D8                                 2161+         DC    A(T2)             TEST &CUR
00002D0C  000013D0                                 2162+         DC    A(T3)             TEST &CUR
00002D10  000014C8                                 2163+         DC    A(T4)             TEST &CUR
00002D14  000015C0                                 2164+         DC    A(T5)             TEST &CUR
00002D18  000016B8                                 2165+         DC    A(T6)             TEST &CUR
00002D1C  000017B0                                 2166+         DC    A(T7)             TEST &CUR
00002D20  000018A8                                 2167+         DC    A(T8)             TEST &CUR
00002D24  000019A0                                 2168+         DC    A(T9)             TEST &CUR
00002D28  00001A98                                 2169+         DC    A(T10)            TEST &CUR
00002D2C  00001B90                                 2170+         DC    A(T11)            TEST &CUR
00002D30  00001C88                                 2171+         DC    A(T12)            TEST &CUR
00002D34  00001D80                                 2172+         DC    A(T13)            TEST &CUR
00002D38  00001E78                                 2173+         DC    A(T14)            TEST &CUR
00002D3C  00001F70                                 2174+         DC    A(T15)            TEST &CUR
00002D40  00002068                                 2175+         DC    A(T16)            TEST &CUR
00002D44  00002160                                 2176+         DC    A(T17)            TEST &CUR
00002D48  00002258                                 2177+         DC    A(T18)            TEST &CUR
00002D4C  00002350                                 2178+         DC    A(T19)            TEST &CUR
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2     STMT

00002D50  00002448                                2179+         DC    A(T20)            TEST &CUR
00002D54  00002540                                2180+         DC    A(T21)            TEST &CUR
00002D58  00002638                                2181+         DC    A(T22)            TEST &CUR
00002D5C  00002730                                2182+         DC    A(T23)            TEST &CUR
00002D60  00002828                                2183+         DC    A(T24)            TEST &CUR
00002D64  00002920                                2184+         DC    A(T25)            TEST &CUR
00002D68  00002A18                                2185+         DC    A(T26)            TEST &CUR
00002D6C  00002B10                                2186+         DC    A(T27)            TEST &CUR
00002D70  00002C08                                2187+         DC    A(T28)            TEST &CUR
                                                  2188+*
00002D74  00000000                                2189+         DC    A(0)              END OF TABLE
00002D78  00000000                                2190+         DC    A(0)
                                                  2191
00002D7C  00000000                                2192          DC    F'0'     END OF TABLE
00002D80  00000000                                2193          DC    F'0'
```

```
  LOC        OBJECT CODE       ADDR1     ADDR2    STMT

                                          2195 ************************************************************************
                                          2196 *        Register equates
                                          2197 ************************************************************************


                             00000000  00000001  2199 R0       EQU   0
                             00000001  00000001  2200 R1       EQU   1
                             00000002  00000001  2201 R2       EQU   2
                             00000003  00000001  2202 R3       EQU   3
                             00000004  00000001  2203 R4       EQU   4
                             00000005  00000001  2204 R5       EQU   5
                             00000006  00000001  2205 R6       EQU   6
                             00000007  00000001  2206 R7       EQU   7
                             00000008  00000001  2207 R8       EQU   8
                             00000009  00000001  2208 R9       EQU   9
                             0000000A  00000001  2209 R10      EQU   10
                             0000000B  00000001  2210 R11      EQU   11
                             0000000C  00000001  2211 R12      EQU   12
                             0000000D  00000001  2212 R13      EQU   13
                             0000000E  00000001  2213 R14      EQU   14
                             0000000F  00000001  2214 R15      EQU   15


                                          2216 ************************************************************************
                                          2217 *        Register equates
                                          2218 ************************************************************************


                             00000000  00000001  2220 FPR0     EQU   0
                             00000001  00000001  2221 FPR1     EQU   1
                             00000002  00000001  2222 FPR2     EQU   2
                             00000003  00000001  2223 FPR3     EQU   3
                             00000004  00000001  2224 FPR4     EQU   4
                             00000005  00000001  2225 FPR5     EQU   5
                             00000006  00000001  2226 FPR6     EQU   6
                             00000007  00000001  2227 FPR7     EQU   7
                             00000008  00000001  2228 FPR8     EQU   8
                             00000009  00000001  2229 FPR9     EQU   9
                             0000000A  00000001  2230 FPR10    EQU   10
                             0000000B  00000001  2231 FPR11    EQU   11
                             0000000C  00000001  2232 FPR12    EQU   12
                             0000000D  00000001  2233 FPR13    EQU   13
                             0000000E  00000001  2234 FPR14    EQU   14
                             0000000F  00000001  2235 FPR15    EQU   15


                                          2237 ************************************************************************
                                          2238 *        Register equates
                                          2239 ************************************************************************


                             00000000  00000001  2241 V0       EQU   0
                             00000001  00000001  2242 V1       EQU   1
                             00000002  00000001  2243 V2       EQU   2
                             00000003  00000001  2244 V3       EQU   3
                             00000004  00000001  2245 V4       EQU   4
```

```
  LOC         OBJECT CODE       ADDR1     ADDR2    STMT

                              00000005  00000001  2246 V5       EQU   5
                              00000006  00000001  2247 V6       EQU   6
                              00000007  00000001  2248 V7       EQU   7
                              00000008  00000001  2249 V8       EQU   8
                              00000009  00000001  2250 V9       EQU   9
                              0000000A  00000001  2251 V10      EQU   10
                              0000000B  00000001  2252 V11      EQU   11
                              0000000C  00000001  2253 V12      EQU   12
                              0000000D  00000001  2254 V13      EQU   13
                              0000000E  00000001  2255 V14      EQU   14
                              0000000F  00000001  2256 V15      EQU   15
                              00000010  00000001  2257 V16      EQU   16
                              00000011  00000001  2258 V17      EQU   17
                              00000012  00000001  2259 V18      EQU   18
                              00000013  00000001  2260 V19      EQU   19
                              00000014  00000001  2261 V20      EQU   20
                              00000015  00000001  2262 V21      EQU   21
                              00000016  00000001  2263 V22      EQU   22
                              00000017  00000001  2264 V23      EQU   23
                              00000018  00000001  2265 V24      EQU   24
                              00000019  00000001  2266 V25      EQU   25
                              0000001A  00000001  2267 V26      EQU   26
                              0000001B  00000001  2268 V27      EQU   27
                              0000001C  00000001  2269 V28      EQU   28
                              0000001D  00000001  2270 V29      EQU   29
                              0000001E  00000001  2271 V30      EQU   30
                              0000001F  00000001  2272 V31      EQU   31
                                                  2273
                                                  2274          END
```

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES												
BEGIN	I	00000200	2	172	136	168	169	170									
CHECKRES	U	00000324	1	252	249												
CHECKVXC	U	00000316	1	247	244												
CTLR0	F	00000724	4	570	182	183	184	185									
DECNUM	C	00001190	16	696	326	328	335	337	342	344	373	375	382	384	389	391	409
					411	418	420	425	427	446	448	455	457	462	464		
DONEXT	U	00000346	1	266	260												
E6TEST	4	00000000	160	710	231												
E6TESTS	F	00002D04	4	2157	224												
EDIT	X	00001164	18	691	327	336	343	374	383	390	410	419	426	447	456	463	
ENDTEST	U	000005F4	1	484	229												
EOJ	I	00000708	4	560	217	487											
EOJPSW	D	000006F8	8	558	560												
FAILCONT	U	000005E4	1	474													
FAILED	F	00001000	4	611	323	354	370	406	443	476	485						
FAILMSG	U	00000574	1	441	264												
FAILPSW	D	00000710	8	562	564												
FAILTEST	I	00000720	4	564	488												
FB0001	F	00000298	8	201	205	206	208										
FLG	X	0000000A	1	717	243	381											
FLG1	X	000011EA	1	868													
FLG10	X	00001AA2	1	1282													
FLG11	X	00001B9A	1	1328													
FLG12	X	00001C92	1	1374													
FLG13	X	00001D8A	1	1420													
FLG14	X	00001E82	1	1468													
FLG15	X	00001F7A	1	1514													
FLG16	X	00002072	1	1560													
FLG17	X	0000216A	1	1606													
FLG18	X	00002262	1	1652													
FLG19	X	0000235A	1	1698													
FLG2	X	000012E2	1	914													
FLG20	X	00002452	1	1744													
FLG21	X	0000254A	1	1790													
FLG22	X	00002642	1	1836													
FLG23	X	0000273A	1	1882													
FLG24	X	00002832	1	1931													
FLG25	X	0000292A	1	1977													
FLG26	X	00002A22	1	2023													
FLG27	X	00002B1A	1	2069													
FLG28	X	00002C12	1	2115													
FLG3	X	000013DA	1	960													
FLG4	X	000014D2	1	1006													
FLG5	X	000015CA	1	1052													
FLG6	X	000016C2	1	1098													
FLG7	X	000017BA	1	1144													
FLG8	X	000018B2	1	1190													
FLG9	X	000019AA	1	1236													
FPCMIE	X	00000740	4	577													
FPCMIO	X	00000734	4	574													
FPCMOFF	X	0000072C	4	572	294	301	891	937	983	1029	1075	1121	1167	1213	1259	1305	1351
					1397	1443	1491	1537	1583	1629	1675	1721	1767	1813	1859	1905	1954
					2000	2046	2092	2138									
FPCMON	X	00000730	4	573													
FPCMOV	X	00000738	4	575													
FPCMSG	U	00000490	1	368	245												

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES				
FPCMUN	X	0000073C	4	576					
FPCPEXP	C	00001052	3	626	385				
FPCPGOT	C	00001067	3	629	392				
FPCPLINE	C	00001008	13	621	631	395			
FPCPLNG	U	00000063	1	631	394				
FPCPNAME	C	00001039	8	624	378				
FPCPNUM	C	00001015	3	622	376				
FPC_R	F	00000040	4	726	243	248	257	278	388	424
FPC_R_1	F	00001220	4	877	895				
FPC_R_10	F	00001AD8	4	1291	1309				
FPC_R_11	F	00001BD0	4	1337	1355				
FPC_R_12	F	00001CC8	4	1383	1401				
FPC_R_13	F	00001DC0	4	1429	1447				
FPC_R_14	F	00001EB8	4	1477	1495				
FPC_R_15	F	00001FB0	4	1523	1541				
FPC_R_16	F	000020A8	4	1569	1587				
FPC_R_17	F	000021A0	4	1615	1633				
FPC_R_18	F	00002298	4	1661	1679				
FPC_R_19	F	00002390	4	1707	1725				
FPC_R_2	F	00001318	4	923	941				
FPC_R_20	F	00002488	4	1753	1771				
FPC_R_21	F	00002580	4	1799	1817				
FPC_R_22	F	00002678	4	1845	1863				
FPC_R_23	F	00002770	4	1891	1909				
FPC_R_24	F	00002868	4	1940	1958				
FPC_R_25	F	00002960	4	1986	2004				
FPC_R_26	F	00002A58	4	2032	2050				
FPC_R_27	F	00002B50	4	2078	2096				
FPC_R_28	F	00002C48	4	2124	2142				
FPC_R_3	F	00001410	4	969	987				
FPC_R_4	F	00001508	4	1015	1033				
FPC_R_5	F	00001600	4	1061	1079				
FPC_R_6	F	000016F8	4	1107	1125				
FPC_R_7	F	000017F0	4	1153	1171				
FPC_R_8	F	000018E8	4	1199	1217				
FPC_R_9	F	000019E0	4	1245	1263				
FPC_XC1	F	00000088	4	733	298	308			
FPC_XC2	F	0000008C	4	734	305	311			
FPC_XC_1	F	00001268	4	884					
FPC_XC_10	F	00001B20	4	1298					
FPC_XC_11	F	00001C18	4	1344					
FPC_XC_12	F	00001D10	4	1390					
FPC_XC_13	F	00001E08	4	1436					
FPC_XC_14	F	00001F00	4	1484					
FPC_XC_15	F	00001FF8	4	1530					
FPC_XC_16	F	000020F0	4	1576					
FPC_XC_17	F	000021E8	4	1622					
FPC_XC_18	F	000022E0	4	1668					
FPC_XC_19	F	000023D8	4	1714					
FPC_XC_2	F	00001360	4	930					
FPC_XC_20	F	000024D0	4	1760					
FPC_XC_21	F	000025C8	4	1806					
FPC_XC_22	F	000026C0	4	1852					
FPC_XC_23	F	000027B8	4	1898					
FPC_XC_24	F	000028B0	4	1947					
FPC_XC_25	F	000029A8	4	1993					

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES												
FPC_XC_26	F	00002AA0	4	2039													
FPC_XC_27	F	00002B98	4	2085													
FPC_XC_28	F	00002C90	4	2131													
FPC_XC_3	F	00001458	4	976													
FPC_XC_4	F	00001550	4	1022													
FPC_XC_5	F	00001648	4	1068													
FPC_XC_6	F	00001740	4	1114													
FPC_XC_7	F	00001838	4	1160													
FPC_XC_8	F	00001930	4	1206													
FPC_XC_9	F	00001A28	4	1252													
FPR0	U	00000000	1	2220													
FPR1	U	00000001	1	2221													
FPR10	U	0000000A	1	2230													
FPR11	U	0000000B	1	2231													
FPR12	U	0000000C	1	2232													
FPR13	U	0000000D	1	2233													
FPR14	U	0000000E	1	2234													
FPR15	U	0000000F	1	2235													
FPR2	U	00000002	1	2222													
FPR3	U	00000003	1	2223													
FPR4	U	00000004	1	2224													
FPR5	U	00000005	1	2225													
FPR6	U	00000006	1	2226													
FPR7	U	00000007	1	2227													
FPR8	U	00000008	1	2228													
FPR9	U	00000009	1	2229													
IMAGE	1	00000000	11652	0													
K	U	00000400	1	594	595	596	597										
K64	U	00010000	1	596													
M4	U	00000008	1	715	334	454											
M5	U	00000009	1	716	341	461											
MB	U	00100000	1	597													
MSG	I	00000638	4	520	216	503											
MSGCMD	C	00000686	9	550	533	534											
MSGMSG	C	0000068F	100	551	527	548	525										
MSGMVC	I	00000680	6	548	531												
MSGOK	I	0000064E	2	529	526												
MSGRET	I	0000066E	4	544	537	540											
MSGSAVE	F	00000674	4	547	523	544											
NEXTE6	U	000002EC	1	226	268	479											
OPNAME	C	00000010	8	720	284	331	378	414	451								
PAGE	U	00001000	1	595													
PRT3	C	0000117A	18	694	327	328	329	336	337	338	343	344	345	374	375	376	383
					384	385	390	391	392	410	411	412	419	420	421	426	427
					428	447	448	449	456	457	458	463	464	465			
PRTLINE	C	000010CE	13	659	669	468											
PRTLNG	U	00000048	1	669	467												
PRTM4	C	00001107	2	664	458												
PRTM5	C	00001113	2	667	465												
PRTNAME	C	000010F6	8	662	451												
PRTNUM	C	000010DB	3	660	449												
R0	U	00000000	1	2199	130	182	185	205	207	208	209	214	233	234	322	323	348
					353	354	369	370	394	405	406	430	442	443	467	475	476
					502	504	520	523	525	527	529	544					
R1	U	00000001	1	2200	215	256	257	258	259	262	263	314	315	349	395	431	468
					485	486	534	548									

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES												
R10	U	0000000A	1	2209	170	179	180										
R11	U	0000000B	1	2210	236	237	897	943	989	1035	1081	1127	1173	1219	1265	1311	1357
					1403	1449	1497	1543	1589	1635	1681	1727	1773	1819	1865	1911	1960
					2006	2052	2098	2144									
R12	U	0000000C	1	2211	224	227	267	478									
R13	U	0000000D	1	2212													
R14	U	0000000E	1	2213													
R15	U	0000000F	1	2214	239	277	279	289	309	312	317	347	350	351	355	396	432
					469	497	507	508									
R2	U	00000002	1	2201	216	325	326	333	334	335	340	341	342	372	373	380	381
					382	387	388	389	408	409	416	417	418	423	424	425	445
					446	453	454	455	460	461	462	502	503	504	521	523	529
					530	531	533	539	544	545	892	893	938	939	984	985	1030
					1031	1076	1077	1122	1123	1168	1169	1214	1215	1260	1261	1306	1307
					1352	1353	1398	1399	1444	1445	1492	1493	1538	1539	1584	1585	1630
					1631	1676	1677	1722	1723	1768	1769	1814	1815	1860	1861	1906	1907
					1955	1956	2001	2002	2047	2048	2093	2094	2139	2140			
R3	U	00000003	1	2202													
R4	U	00000004	1	2203													
R5	U	00000005	1	2204	227	228	231	498	506	861	899	907	945	953	991	999	1037
					1045	1083	1091	1129	1137	1175	1183	1221	1229	1267	1275	1313	1321
					1359	1367	1405	1413	1451	1461	1499	1507	1545	1553	1591	1599	1637
					1645	1683	1691	1729	1737	1775	1783	1821	1829	1867	1875	1913	1924
					1962	1970	2008	2016	2054	2062	2100	2108	2146				
R6	U	00000006	1	2205													
R7	U	00000007	1	2206	245	250	397	433									
R8	U	00000008	1	2207	168	172	173	174	176								
R9	U	00000009	1	2208	169	176	177	179									
RE1	F	000012A4	4	898	870	873											
RE10	F	00001B5C	4	1312	1284	1287											
RE11	F	00001C54	4	1358	1330	1333											
RE12	F	00001D4C	4	1404	1376	1379											
RE13	F	00001E44	4	1450	1422	1425											
RE14	F	00001F3C	4	1498	1470	1473											
RE15	F	00002034	4	1544	1516	1519											
RE16	F	0000212C	4	1590	1562	1565											
RE17	F	00002224	4	1636	1608	1611											
RE18	F	0000231C	4	1682	1654	1657											
RE19	F	00002414	4	1728	1700	1703											
RE2	F	0000139C	4	944	916	919											
RE20	F	0000250C	4	1774	1746	1749											
RE21	F	00002604	4	1820	1792	1795											
RE22	F	000026FC	4	1866	1838	1841											
RE23	F	000027F4	4	1912	1884	1887											
RE24	F	000028EC	4	1961	1933	1936											
RE25	F	000029E4	4	2007	1979	1982											
RE26	F	00002ADC	4	2053	2025	2028											
RE27	F	00002BD4	4	2099	2071	2074											
RE28	F	00002CCC	4	2145	2117	2120											
RE3	F	00001494	4	990	962	965											
RE4	F	0000158C	4	1036	1008	1011											
RE5	F	00001684	4	1082	1054	1057											
RE6	F	0000177C	4	1128	1100	1103											
RE7	F	00001874	4	1174	1146	1149											
RE8	F	0000196C	4	1220	1192	1195											
RE9	F	00001A64	4	1266	1238	1241											

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES				
READDR	A	0000001C	4	722	262				
REG2LOW	U	000000DD	1	601					
REG2PATT	U	AABBCCDD	1	600					
RELEN	A	00000018	4	721					
RPTDWSAV	D	00000628	8	513	502	504			
RPTERROR	I	00000602	4	497	350	396	432	469	
RPTSAVE	F	00000620	4	510	497	507			
RPTSVR5	F	00000624	4	511	498	506			
SKIPXC	C	00000050	8	728	275	283	288		
SKL0001	U	0000005F	1	198	214				
SKT0001	C	0000022A	26	195	198	215			
SVOLDPSW	U	00000140	0	132					
T1	A	000011E0	4	862	2160				
T10	A	00001A98	4	1276	2169				
T11	A	00001B90	4	1322	2170				
T12	A	00001C88	4	1368	2171				
T13	A	00001D80	4	1414	2172				
T14	A	00001E78	4	1462	2173				
T15	A	00001F70	4	1508	2174				
T16	A	00002068	4	1554	2175				
T17	A	00002160	4	1600	2176				
T18	A	00002258	4	1646	2177				
T19	A	00002350	4	1692	2178				
T2	A	000012D8	4	908	2161				
T20	A	00002448	4	1738	2179				
T21	A	00002540	4	1784	2180				
T22	A	00002638	4	1830	2181				
T23	A	00002730	4	1876	2182				
T24	A	00002828	4	1925	2183				
T25	A	00002920	4	1971	2184				
T26	A	00002A18	4	2017	2185				
T27	A	00002B10	4	2063	2186				
T28	A	00002C08	4	2109	2187				
T3	A	000013D0	4	954	2162				
T4	A	000014C8	4	1000	2163				
T5	A	000015C0	4	1046	2164				
T6	A	000016B8	4	1092	2165				
T7	A	000017B0	4	1138	2166				
T8	A	000018A8	4	1184	2167				
T9	A	000019A0	4	1230	2168				
TESTING	F	00001004	4	612	234				
TNUM	H	00000004	2	712	233	325	372	408	445
TSUB	A	00000000	4	711	236				
TTABLE	F	00002D04	4	2159					
V0	U	00000000	1	2241					
V1	U	00000001	1	2242					
V10	U	0000000A	1	2251					
V11	U	0000000B	1	2252					
V12	U	0000000C	1	2253					
V13	U	0000000D	1	2254					
V14	U	0000000E	1	2255					
V15	U	0000000F	1	2256	297	299	304	306	
V16	U	00000010	1	2257	296	297	303	304	
V17	U	00000011	1	2258					
V18	U	00000012	1	2259					
V19	U	00000013	1	2260					

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
V1FUDGE	X	000011B0	16	702	
V1INPUT	X	000011C0	16	703	
V1O1	X	00001208	16	875	896
V1O10	X	00001AC0	16	1289	1310
V1O11	X	00001BB8	16	1335	1356
V1O12	X	00001CB0	16	1381	1402
V1O13	X	00001DA8	16	1427	1448
V1O14	X	00001EA0	16	1475	1496
V1O15	X	00001F98	16	1521	1542
V1O16	X	00002090	16	1567	1588
V1O17	X	00002188	16	1613	1634
V1O18	X	00002280	16	1659	1680
V1O19	X	00002378	16	1705	1726
V1O2	X	00001300	16	921	942
V1O20	X	00002470	16	1751	1772
V1O21	X	00002568	16	1797	1818
V1O22	X	00002660	16	1843	1864
V1O23	X	00002758	16	1889	1910
V1O24	X	00002850	16	1938	1959
V1O25	X	00002948	16	1984	2005
V1O26	X	00002A40	16	2030	2051
V1O27	X	00002B38	16	2076	2097
V1O28	X	00002C30	16	2122	2143
V1O3	X	000013F8	16	967	988
V1O4	X	000014F0	16	1013	1034
V1O5	X	000015E8	16	1059	1080
V1O6	X	000016E0	16	1105	1126
V1O7	X	000017D8	16	1151	1172
V1O8	X	000018D0	16	1197	1218
V1O9	X	000019C8	16	1243	1264
V1OUTPUT	X	00000028	16	724	263 296 303
V2	U	00000002	1	2243	
V20	U	00000014	1	2261	
V21	U	00000015	1	2262	
V22	U	00000016	1	2263	893 894 896 939 940 942 985 986 988 1031 1032 1034 1077 1078 1080 1123 1124 1126 1169 1170 1172 1215 1216 1218 1261 1262 1264 1307 1308 1310 1353 1354 1356 1399 1400 1402 1445 1446 1448 1493 1494 1496 1539 1540 1542 1585 1586 1588 1631 1632 1634 1677 1678 1680 1723 1724 1726 1769 1770 1772 1815 1816 1818 1861 1862 1864 1907 1908 1910 1956 1957 1959 2002 2003 2005 2048 2049 2051 2094 2095 2097 2140 2141 2143
V23	U	00000017	1	2264	893 894 939 940 985 986 1031 1032 1077 1078 1123 1124 1169 1170 1215 1216 1261 1262 1307 1308 1353 1354 1399 1400 1445 1446 1493 1494 1539 1540 1585 1586 1631 1632 1677 1678 1723 1724 1769 1770 1815 1816 1861 1862 1907 1908 1956 1957 2002 2003 2048 2049 2094 2095 2140 2141
V24	U	00000018	1	2265	
V25	U	00000019	1	2266	
V26	U	0000001A	1	2267	
V27	U	0000001B	1	2268	
V28	U	0000001C	1	2269	
V29	U	0000001D	1	2270	
V2ADDR	A	0000000C	4	719	314
V2_1	A	000011EC	4	870	892
V2_10	A	00001AA4	4	1284	1306
V2_11	A	00001B9C	4	1330	1352

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES
V2_12	A	00001C94	4	1376	1398
V2_13	A	00001D8C	4	1422	1444
V2_14	A	00001E84	4	1470	1492
V2_15	A	00001F7C	4	1516	1538
V2_16	A	00002074	4	1562	1584
V2_17	A	0000216C	4	1608	1630
V2_18	A	00002264	4	1654	1676
V2_19	A	0000235C	4	1700	1722
V2_2	A	000012E4	4	916	938
V2_20	A	00002454	4	1746	1768
V2_21	A	0000254C	4	1792	1814
V2_22	A	00002644	4	1838	1860
V2_23	A	0000273C	4	1884	1906
V2_24	A	00002834	4	1933	1955
V2_25	A	0000292C	4	1979	2001
V2_26	A	00002A24	4	2025	2047
V2_27	A	00002B1C	4	2071	2093
V2_28	A	00002C14	4	2117	2139
V2_3	A	000013DC	4	962	984
V2_4	A	000014D4	4	1008	1030
V2_5	A	000015CC	4	1054	1076
V2_6	A	000016C4	4	1100	1122
V2_7	A	000017BC	4	1146	1168
V2_8	A	000018B4	4	1192	1214
V2_9	A	000019AC	4	1238	1260
V3	U	00000003	1	2244	
V30	U	0000001E	1	2271	
V31	U	0000001F	1	2272	
V4	U	00000004	1	2245	
V5	U	00000005	1	2246	
V6	U	00000006	1	2247	
V7	U	00000007	1	2248	
V8	U	00000008	1	2249	
V9	U	00000009	1	2250	
VXC	X	0000000B	1	718	248 417
VXC1	X	000011EB	1	869	
VXC10	X	00001AA3	1	1283	
VXC11	X	00001B9B	1	1329	
VXC12	X	00001C93	1	1375	
VXC13	X	00001D8B	1	1421	
VXC14	X	00001E83	1	1469	
VXC15	X	00001F7B	1	1515	
VXC16	X	00002073	1	1561	
VXC17	X	0000216B	1	1607	
VXC18	X	00002263	1	1653	
VXC19	X	0000235B	1	1699	
VXC2	X	000012E3	1	915	
VXC20	X	00002453	1	1745	
VXC21	X	0000254B	1	1791	
VXC22	X	00002643	1	1837	
VXC23	X	0000273B	1	1883	
VXC24	X	00002833	1	1932	
VXC25	X	0000292B	1	1978	
VXC26	X	00002A23	1	2024	
VXC27	X	00002B1B	1	2070	
VXC28	X	00002C13	1	2116	

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES	
VXC3	X	000013DB	1	961		
VXC4	X	000014D3	1	1007		
VXC5	X	000015CB	1	1053		
VXC6	X	000016C3	1	1099		
VXC7	X	000017BB	1	1145		
VXC8	X	000018B3	1	1191		
VXC9	X	000019AB	1	1237		
VXCMSG	U	00000502	1	404	250	
VXCPEXP	C	000010B5	3	645	421	
VXCPGOT	C	000010CA	3	648	428	
VXCPLINE	C	0000106B	13	640	650	431
VXCPLNG	U	00000063	1	650	430	
VXCPNAME	C	0000109C	8	643	414	
VXCPNUM	C	00001078	3	641	412	
WK1	F	00000098	4	736		
WK2	F	0000009C	4	737		
X0001	U	000002C0	1	204	192	205
X1	F	00001280	4	890	862	
X10	F	00001B38	4	1304	1276	
X11	F	00001C30	4	1350	1322	
X12	F	00001D28	4	1396	1368	
X13	F	00001E20	4	1442	1414	
X14	F	00001F18	4	1490	1462	
X15	F	00002010	4	1536	1508	
X16	F	00002108	4	1582	1554	
X17	F	00002200	4	1628	1600	
X18	F	000022F8	4	1674	1646	
X19	F	000023F0	4	1720	1692	
X2	F	00001378	4	936	908	
X20	F	000024E8	4	1766	1738	
X21	F	000025E0	4	1812	1784	
X22	F	000026D8	4	1858	1830	
X23	F	000027D0	4	1904	1876	
X24	F	000028C8	4	1953	1925	
X25	F	000029C0	4	1999	1971	
X26	F	00002AB8	4	2045	2017	
X27	F	00002BB0	4	2091	2063	
X28	F	00002CA8	4	2137	2109	
X3	F	00001470	4	982	954	
X4	F	00001568	4	1028	1000	
X5	F	00001660	4	1074	1046	
X6	F	00001758	4	1120	1092	
X7	F	00001850	4	1166	1138	
X8	F	00001948	4	1212	1184	
X9	F	00001A40	4	1258	1230	
XC0001	U	000002E8	1	218	210	
XCFAILMSG	H	000003D6	2	321	316	
XCHECK	U	0000034E	1	274	239	
XCO1	X	00001240	16	881		
XCO10	X	00001AF8	16	1295		
XCO11	X	00001BF0	16	1341		
XCO12	X	00001CE8	16	1387		
XCO13	X	00001DE0	16	1433		
XCO14	X	00001ED8	16	1481		
XCO15	X	00001FD0	16	1527		
XCO16	X	000020C8	16	1573		

SYMBOL	TYPE	VALUE	LENGTH	DEFN	REFERENCES				
XCO17	X	000021C0	16	1619					
XCO18	X	000022B8	16	1665					
XCO19	X	000023B0	16	1711					
XCO2	X	00001338	16	927					
XCO20	X	000024A8	16	1757					
XCO21	X	000025A0	16	1803					
XCO22	X	00002698	16	1849					
XCO23	X	00002790	16	1895					
XCO24	X	00002888	16	1944					
XCO25	X	00002980	16	1990					
XCO26	X	00002A78	16	2036					
XCO27	X	00002B70	16	2082					
XCO28	X	00002C68	16	2128					
XCO3	X	00001430	16	973					
XCO4	X	00001528	16	1019					
XCO5	X	00001620	16	1065					
XCO6	X	00001718	16	1111					
XCO7	X	00001810	16	1157					
XCO8	X	00001908	16	1203					
XCO9	X	00001A00	16	1249					
XCOUTPUT	X	00000060	16	730	299	306	315		
XCPLINE	C	00001116	13	677	686	349			
XCPLNG	U	0000004E	1	686	348				
XCPM4	C	00001156	2	682	338				
XCPM5	C	00001161	2	684	345				
XCPNAME	C	00001145	8	680	331				
XCPTNUM	C	00001123	3	678	329				
XCR15	F	00000488	8	361	347	351			
XCRESULT	X	00000458	16	358					
XCSKIP	C	00000007	1	714	276				
XCV1	X	00000468	16	359					
XCV2	X	00000478	16	360					
XCVCRNF	F	00000380	4	293	285				
ZVE6TST	J	00000000	11652	129	132	134	138	142	610	130
=A(E6TESTS)	A	00000764	4	585	224				
=AL2(L'MSGMSG)	R	00000772	2	589	525				
=CL1'S'	C	00000774	1	590	276				
=CL8' '	C	00000750	8	582	283				
=CL8'SKIP XC '	C	00000748	8	581	275	288			
=CL8'VCRNF'	C	00000758	8	583	284				
=F'1'	F	0000076C	4	587	322	353	369	405	442	475
=F'4'	F	00000760	4	584	209				
=H'0'	H	00000770	2	588	520				
=XL4'00000008'	X	00000768	4	586	258				

```
 MACRO   DEFN  REFERENCES

FCHECK     81    191
PTTABLE   817   2158
VRR_C     753    859    905    951    997   1043   1089   1135   1181   1227   1273   1319   1365   1411   1459   1505   1551   1597
               1643   1689   1735   1781   1827   1873   1922   1968   2014   2060   2106
```

```
   DESC     SYMBOL   SIZE     POS        ADDR

Entry: 0

Image      IMAGE    11652  0000-2D83  0000-2D83
  Region            11652  0000-2D83  0000-2D83
    CSECT  ZVE6TST  11652  0000-2D83  0000-2D83
```

```
   STMT                      FILE NAME
1     /devstor/sharedvfp/tests/zvector-e6-21-VCRNF.asm

** NO ERRORS FOUND **
```